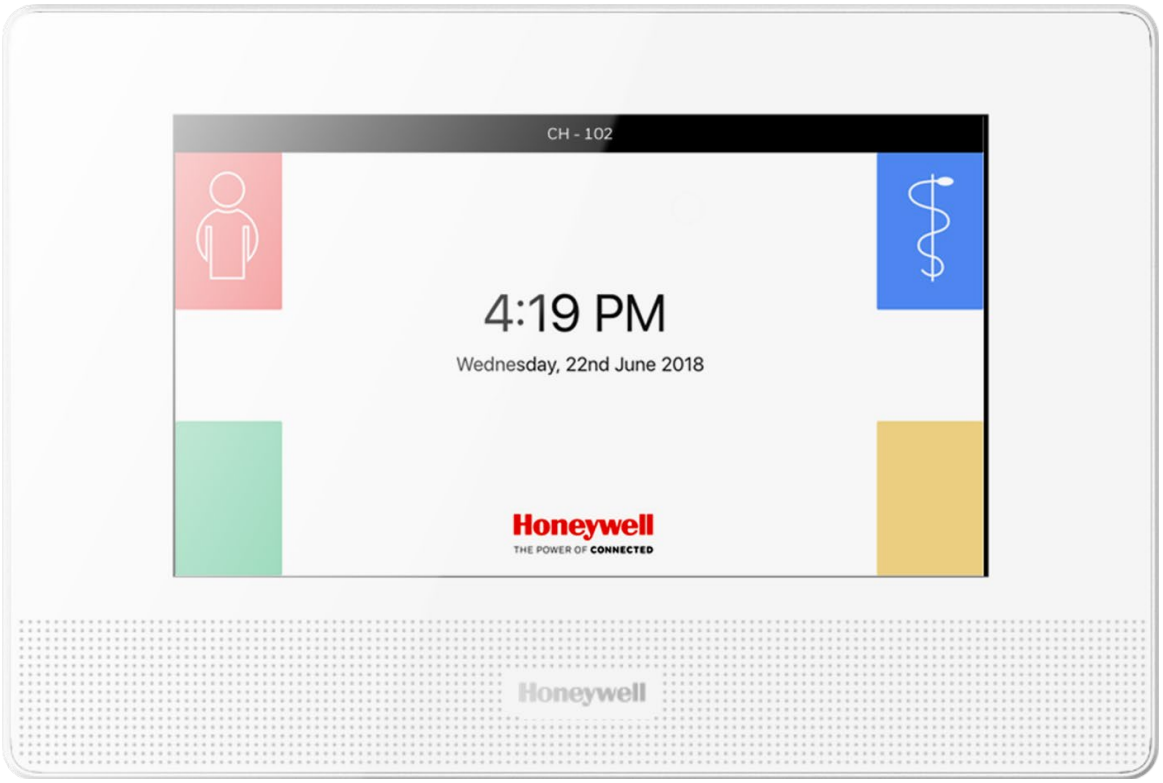
Honeywell | Intelligent Life Care



Systevo Touch IP (STIP)

GB Installation Instruction

89470J4.GB0
08.2022

Intended purpose
This products may only be used for the applications outlined in the catalogue and in the technical description, and only in conjunction with the recommended and approved external devices and components.
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Safety-related user information
This manual includes all information required for the proper use of the products described.
In order to ensure correct and safe operation of the product, all guidelines concerning its transport, storage, installation, and mounting must be observed. This includes the necessary care in operating the product.
The term 'qualified personnel' in the context of the safety information included in this manual or on the product itself designates:

- project engineers who are familiar with the safety guidelines concerning patient call systems incl. associated components.
- trained service engineers who are familiar with the components of patient call systems and the information on their operation as included in this manual.
- trained installation or service personnel with the necessary qualification for carrying out repairs on patient call systems incl. associated components or who are authorised to operate, ground and label electrical circuits and/or safety equipment/systems.

Symbols
The following information is given in the interest of personal safety and to prevent damage to the product described in this manual and all equipment connected to it.
Safety information and warnings for the prevention of dangers putting at risk the life and health of user and maintenance personnel as well as causing damage to the equipment itself are marked by the following pictograms. Within the context of this manual, these pictograms have the following meanings:



Warning - Designates risks for man and/or machine. Non-compliance will create risks to man and/or machine. The level of risk is indicated by the word of warning.



Note - Important information on a topic or a procedure and other important information!



Standards and guidelines - Observe configuration and commissioning information in accordance to the national and local requirements.

Demontage



In accordance with Directive 2002/96/EG (WEEE), after being dismantled, electrical and electronic equipment is taken back by the manufacturer for proper disposal.

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1 General / Application

The Honeywell Systevo Touch IP terminal, hereinafter **STIP**, is a modern communication device equipped with a Touch terminal for care facilities and consists of a control unit (Touch terminal) and a connection board for connecting to the corridor data bus and all corridor components. Additional network interfaces enable a connection to the call system and possible additional services. The connection board serves as an interface between the different communication levels and as a central control unit for the functions in the corridor.

Additional and current information

The described features, specifications and product related information in this manual correspond to the date of issue (refer to date on the front page) and may differ due to modifications and/or amended Standards and Regulations of the System design, Installation and Commissioning.

Updated information and declaration of conformity are available for comparison on the www.ackermann-clino.com homepage.

Safety information:

- All mounting and installation work must be carried out only when the device is disconnected from the power.
- In accordance with DIN VDE 0834-1, connecting cables (including cable laying) must be able to withstand a test voltage of 4 KV. The voltage supply of the devices can be fed in the same cable. Observe the appropriate wire cross-sections! The 89734AH / 89734AJ system cable is recommend for the corridor data bus cabling.
- When using PoE as the power supply, it is not possible to establish safe isolation with 2 x MOPP in accordance with DIN EN 60601-1, as stipulated in DIN VDE 0834-1:2016-06.
- If required, use a network isolator in accordance with EN60601-1 (Part no. 79823D1).
- If inductive loads are connected, a free-wheeling diode absolutely must be used. Otherwise, the switch outputs may get damaged.
- Devices which are external to the system should be connected only via interfaces which ensure safe isolation from dangerous shock currents in accordance with DIN EN 60601-1 (e.g. power surge relay for relay contacts)
- For 24 V DC power supply from the ward, a standard network connection without PoE must be used.
- A functional test is mandatory to ensure safe operation in accordance with current regulations (DIN VDE 0834-1:2016-06).

Notice:
The STIP only supports the DIN standard VDE 0834-1: 2000-04 and DIN standard VDE 0834-1: 2016-06.

1.1 Other applicable documentation

Name	Part No.
Technical information Systevo Call Ackermann with IP devices	89470F8
Commissioning Instruction Systevo Touch Terminal IP (STIP)	89470J7.GB0
Installation Instruction Systevo Call Ackermann	89470A5
Installation Instruction Systevo Call Ackermann Systevo Control IP	89470A6
Installation Instruction Systevo bed unit	89470K1
Installation Instruction Systevo Electronic Module SEM & SEM+ / Corridor Lamp SCL	89470K2

1.2 List of abbreviations

Abbreviation	Name	Abbreviation	Name
ABUS	Audio bus	RTWC	Call button bathroom WC
R-ABUS	Corridor audio data bus	RLWC	Call lamp bathroom WC
BBUS	Bed bus	ATWC	Cancel button bathroom/WC call
DBUS	Data bus	AL	Presence lamp
ZBUS	Corridor data bus	STIP	Systevo Touch IP terminal
I/O port	Input / output	SCL	Systevo corridor lamp
ID	Identification on the data bus	SAS	Systevo Audio Switch
PoE	Power over Ethernet	SCU	Systevo Care Unit
Eth	Ethernet	SBU-DM2+	Systevo bed unit DM2+
LAN	Local area network	SBU-D2+	Systevo bed unit D2+
WIFI	Wireless local area network	SBU-M5	Systevo bed unit M5
RT	Call button	SBU-M2	Systevo bed unit M2
RL	Call lamp	SAI	Systevo Audio Interface
BL	Reassurance lamp		

1.3 Requirements

- **System software:**

  In order to use the STIP, the system software must be updated to version V12.xx or above. The system software is continuously being upgraded with new features. The following overview reflects the relationship between system software versions and the possible range of functions at the time of printing. The system software is subject to continuous further development; the software must always be updated to the latest version so that all available functions can be used. The latest software and up-to-date documentation can be found in the protected download area at www.Ackermann-clino.de

System software	Functions
From V12.01	• **Hybrid mode**: Connection to call system via LAN* • **IP mode**: Connection to call system via LAN • Pre-defined **I/O profiles** (1+2)
From V12.02	• **Hybrid mode**: Connection to call system via LAN* • **IP mode**: Connection to call system via LAN • Pre-defined I/O profiles (1+2+3)
From V12.03	• **Data bus mode**: Connection to call system via ZBUS
From V12.04	• **IP+ mode**: Connection to call system via ZBUS & LAN

  *An additional e-module is required to connect to the nurse call.

- **LAN:**

  A LAN network is set up according to the Ethernet standard (see documentation “Systevo Call Ackermann with IP devices" 89470F8).

2 Installation

Installation information

- All mounting and installation work must be carried out only when the device is disconnected from the power.
- The connecting terminal for the power supply from the corridor data bus is suitable for two cables per pole with a cable cross-section of up to 1.5 mm².
- All other connecting terminals are suitable for installation cables with a wire diameter of 0.6 mm / 0.8 mm.
- Up to three wires measuring 0.6 mm or up to two wires measuring 0.8 mm can be safely connected per screw terminal. For cabling in the room, a wire diameter of 0.6 mm is recommended.
- Strip wires to approx. 5 mm.
- The connection is established via a patch cable on the back of the STIP housing to connect to a data network.
- Allow sufficient installation space for the data connection, e.g. with a flush tunnel box (see Chapter 2.3).
- When installing on the wall, use suitable screws with a maximum screw head height of 3 mm for the screw fitting.

2.1 Mounting

Dimensions

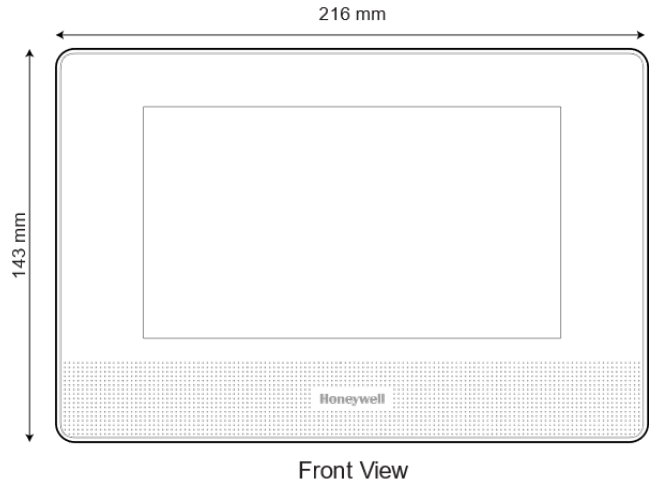


Fig. 1: Dimensions / front view

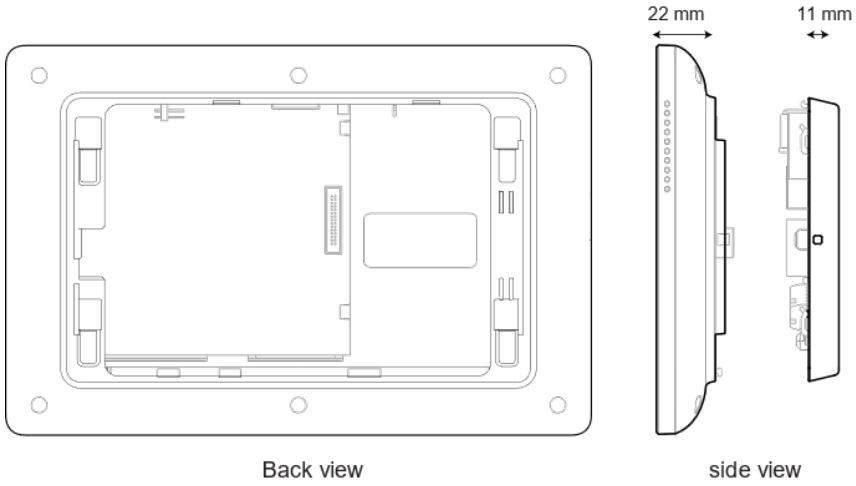


Fig. 2: Dimensions / back view and side view

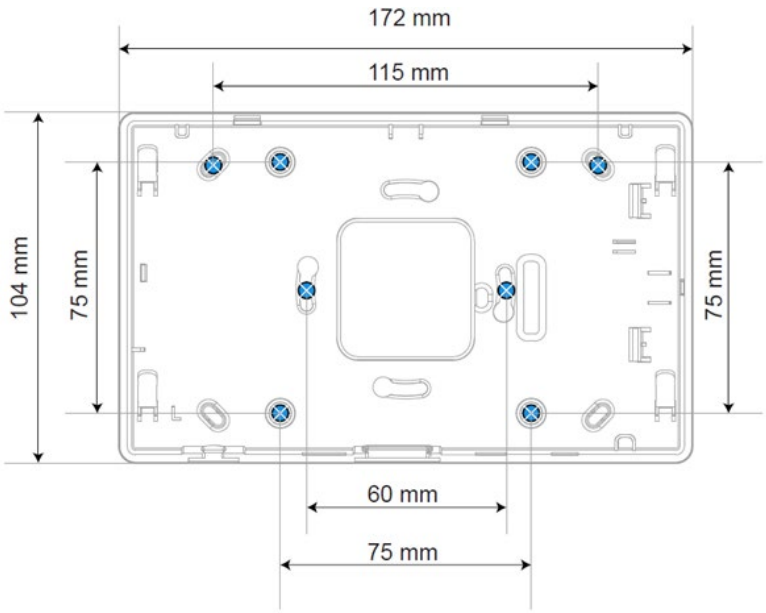
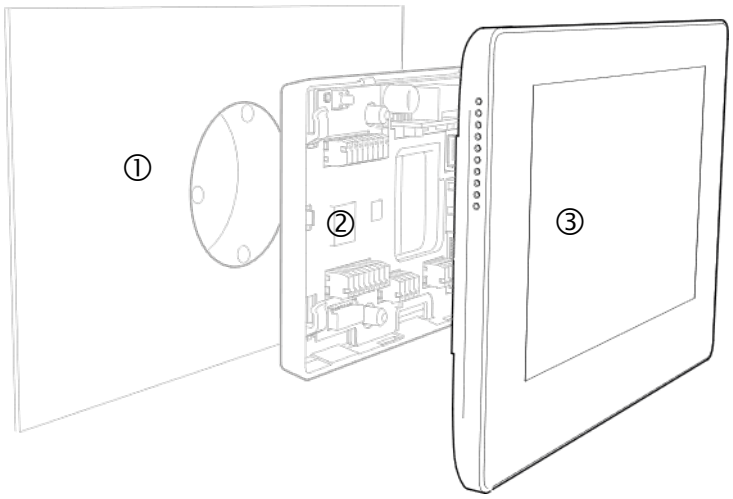


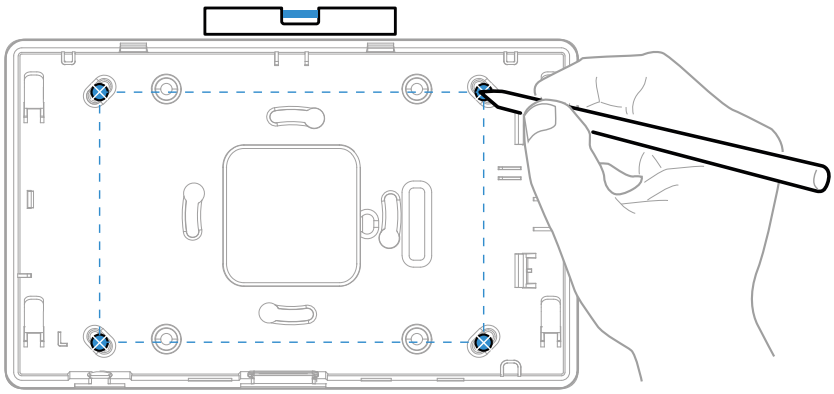
Fig. 3: Dimensions of connection board

2.2 Installation on a wall



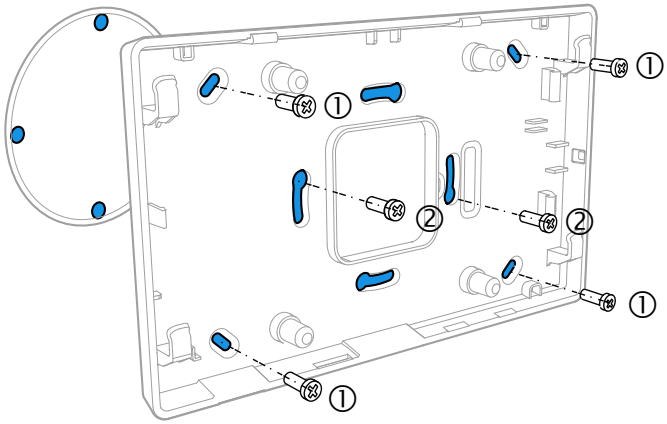
①	Wall
②	Connection board
③	Touch terminal

Fig. 4: Mounting on the wall



Hold the connection board's bottom housing panel horizontally (align using a spirit level) against the mounting surface and mark the four mounting holes.
Drill through the marks for the holes to create four holes and insert wall plugs measuring 5 / 6 mm.

Fig. 5: Mark and drill the mounting holes



Hold the bottom housing panel against the mounting surface and align. Using the four screws ① with a diameter of 4 mm and a max. screw head height of 3 mm, screw in the wall plugs and tighten.
Screw in the two screws ② to fasten on a standard flush box (e.g. tunnel socket) and tighten.
The suitable type of wall plugs and screws depends on the material of the mounting surface (cavity wall, concrete, wood etc.) and must be chosen accordingly.

Fig. 6: Fitting the panel to the mounting surface

The maximum screw head height of 3 mm must be observed, otherwise the STIP electronics could be damaged.

2.3 Mounting on installation socket (e.g. tunnel socket)

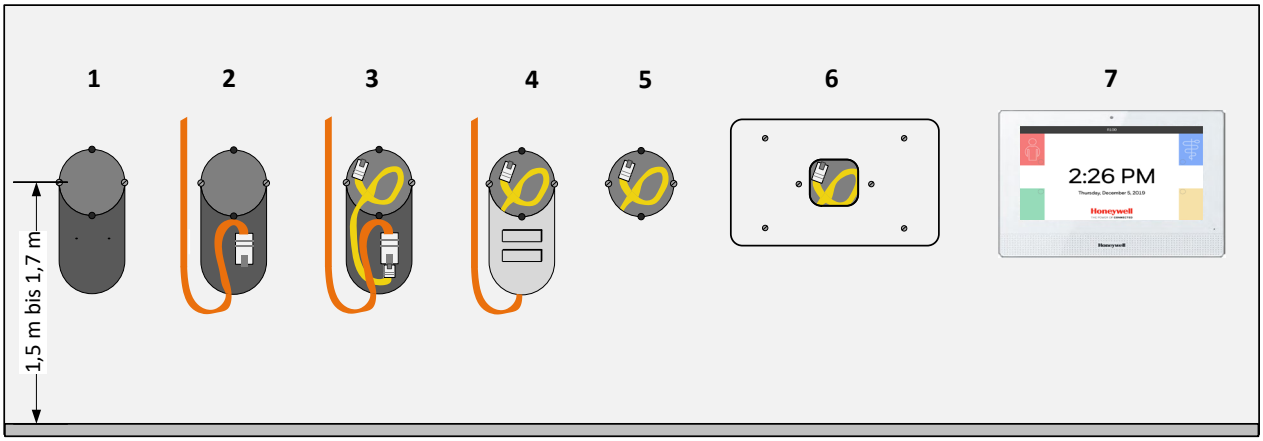


Fig. 7: Keep the mounting height 1.5 m to 1.7 m over the operator's standing surface in accordance with DIN VDE 0834

Installation steps	
1	Flush tunnel box embedded in the wall.
2	CAT cable inserted and data connection equipment with RJ45 Ethernet socket.
3	Use the patch cable included with the connection board.
4	View with closed tunnel.
5	CAT cable and tunnel concealed.
6	Installation of the connection board on the tunnel socket.
7	Connect and install the Touch terminal.

In practice, the "tunnel" of the installation socket is plastered over and thus not visible.

2.4 Inserting the connection board into the bottom housing panel

The connection board features a mechanical catch with a tamper contact to prevent unauthorised removal of the Touch terminal.

Observe the catch position before installation.

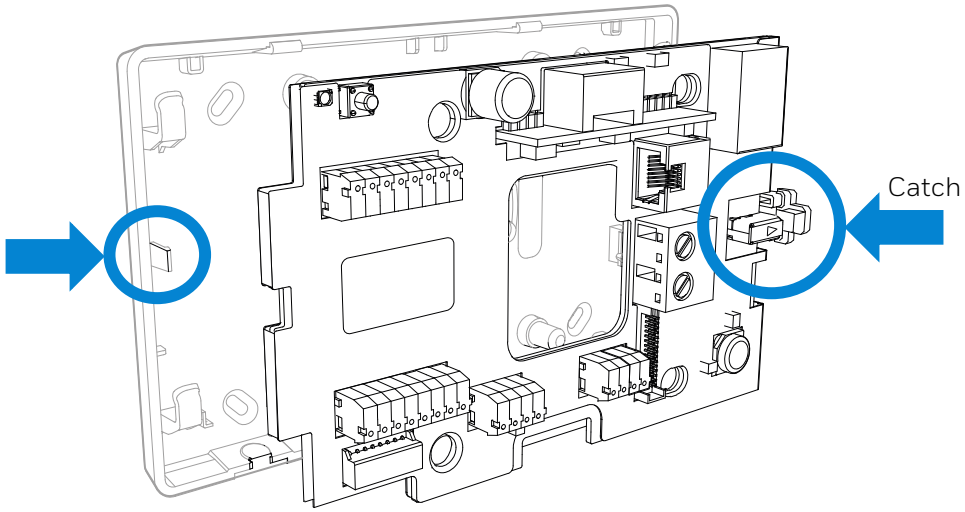


Fig. 8: Mechanical catch / tamper contact

Before inserting the connection board into the bottom housing panel, make sure that the mechanical catch / tamper contact is **pressed** down and set to the “Unlock level” position.

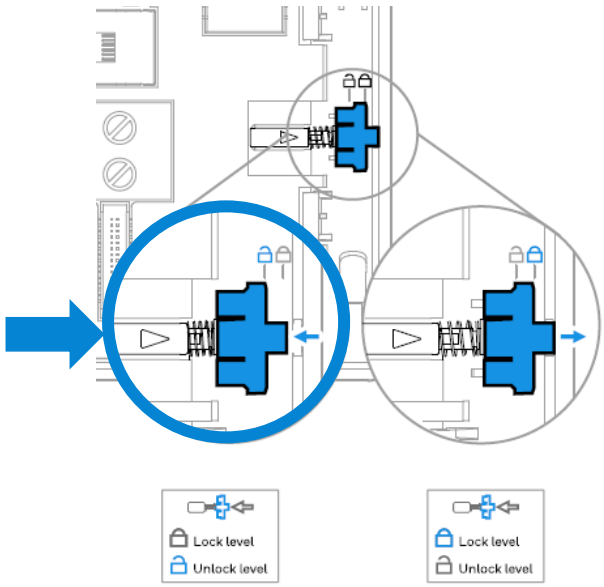


Fig. 9: Position ➔ Unlock level

Inserting the connection board into the bottom housing panel.
Insert the connection board into the lateral guides on the right side of the bottom housing panel.

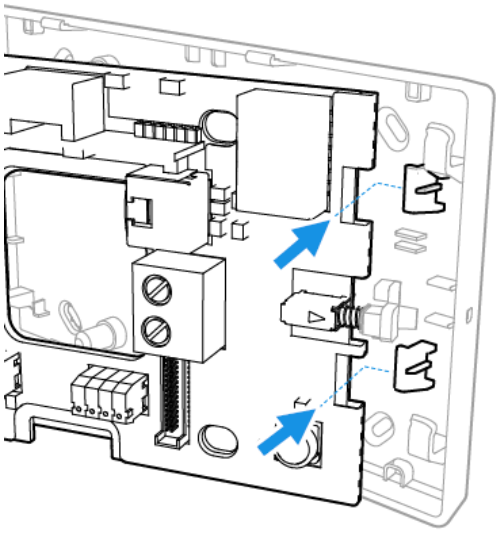


Fig. 10: Lateral guides ➔ right

Then carefully press down the left side of the connection board until it engages.

Snap lock

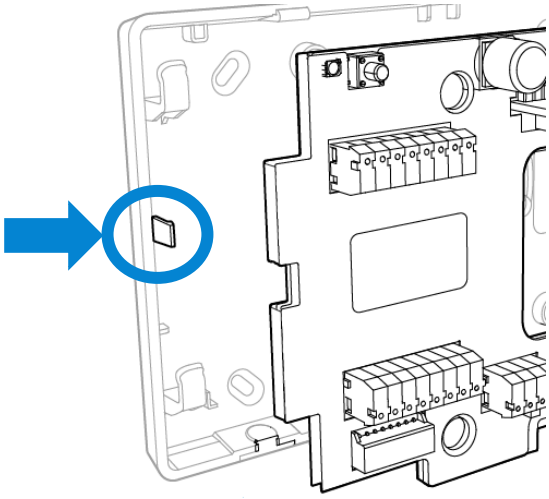


Fig. 11: Snap lock ➔ left

2.5 Installation of the Touch terminal and connection board

Connect the Touch terminal and the connection board to the power supply and install.

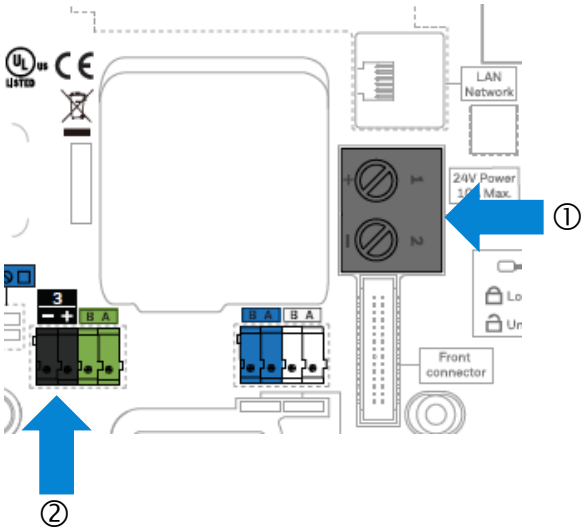


Fig. 12: Position of the connecting terminals for the power supply

①	Power supply for the connection board 79CM307 Z1 / Z2 / Z3 via connecting terminals 1 + and 2 - depending on the version.
②	Power supply for the connection board 79CM307 Z4 also possible via connecting terminal 3 - / +.

Damage hazard!
Assembly and installation work may only be performed while the devices are **de-energised (without power)**.

Attach the connection plug ③ into the “front connector” ④ socket of the connection board in the correct position (an incorrect connection (reverse polarity) is not possible due to the coding lug on the plug and socket). Then re-insert the plug for the power supply / Ethernet (PoE).

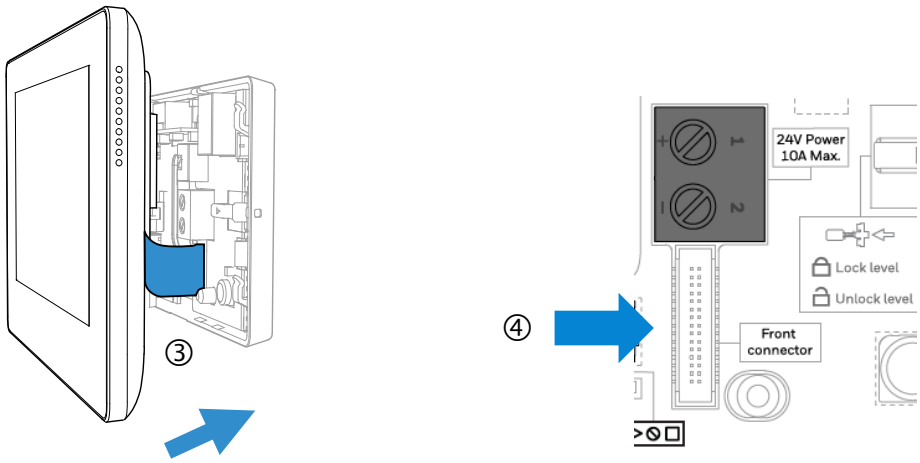


Fig. 13: Connection plug and socket “front connector”

The bottom housing panel has four catch hooks ⑤, (marked in blue) where the Touch terminal engages.

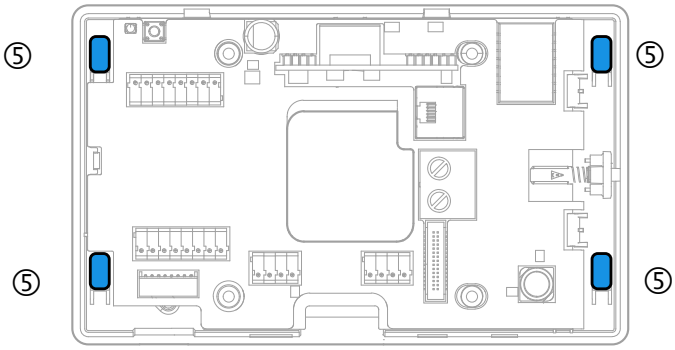


Fig. 14: Position of the catch hooks

Before placing the Touch terminal on the connection board, the electro-mechanical catch must be set to the “Unlock level” ⑥ state, as otherwise the mechanism could be damaged.

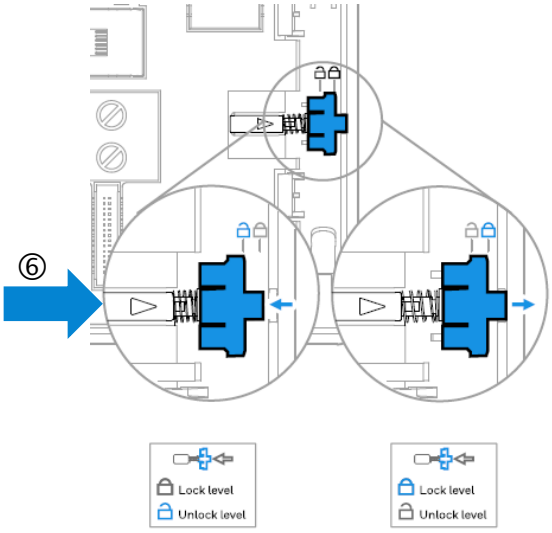
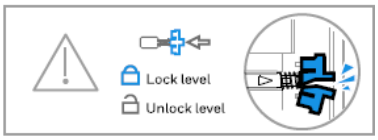


Fig. 15: Position ➔ Unlock level



mechanism damaged

In the “Lock level” state, the Touch terminal cannot be fitted and locked.

Non-observance could cause the mechanism to become damaged.

Place the Touch terminal on the bottom housing panel from above so that the protrusions on the back of the Touch terminal engage in the catch hooks ⑦. Then press the Touch terminal down using moderate pressure until it audibly clicks into place.

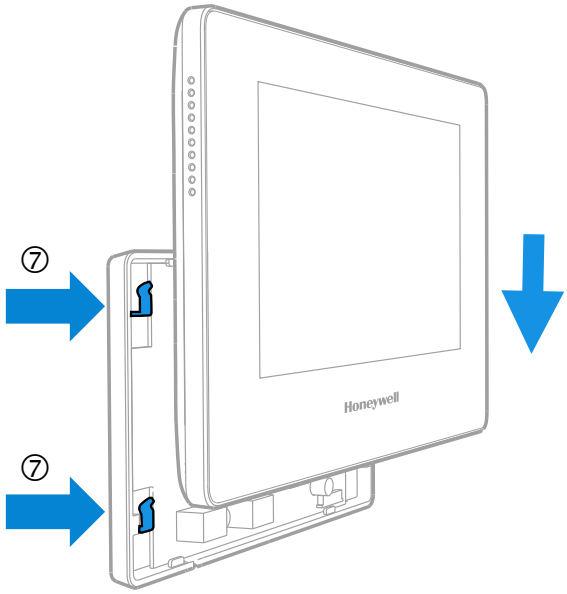


Fig. 16: Place the Touch terminal on the bottom housing panel from above

The electro-mechanical catch will automatically be set to the “Lock level” state ⑧ and thus lock the mechanism.

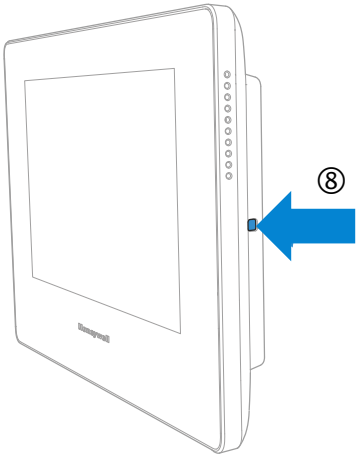


Fig. 17: State ➔ Lock level

In this state, the Touch terminal cannot be removed from the bottom housing panel.

2.6 Removing the Touch terminal from the bottom housing panel / connection board

The Touch terminal can be removed from the connection board for servicing purposes.
If you wish to remove the Touch terminal, first carry out the “Deregister the device” maintenance function* in the service menu.
*This function will become available at a later date (> V12.04).

Then use a thin screwdriver with an edge/tip measuring < 3 mm or similar to unlatch the mechanical catch through the housing opening (Unlock level).

After unlocking, the Touch terminal can be removed from the bottom housing panel by pushing it upward.

The mechanical catch will then be automatically be reset to the “Lock level” position. **Before** replacing the Touch terminal in the bottom housing panel, make sure that the mechanical catch is **pressed** down and set to the “Unlock level” position.

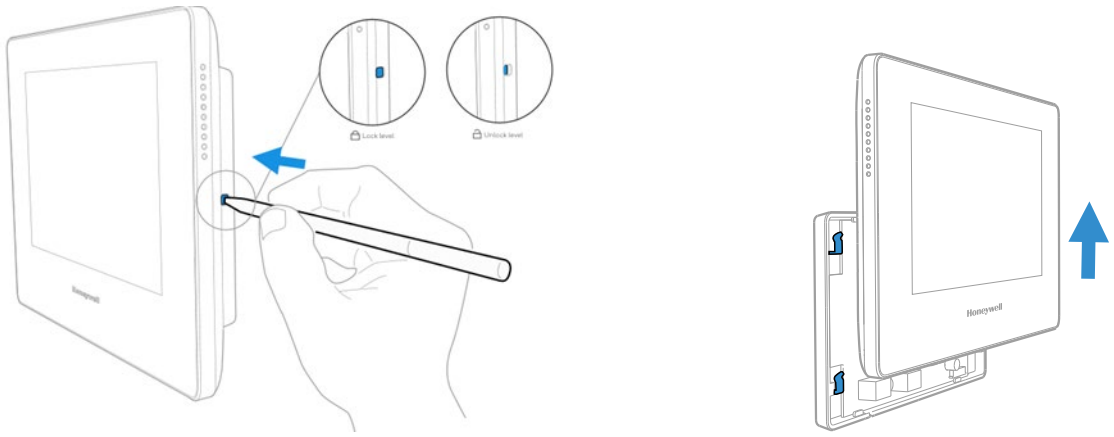


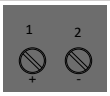
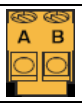
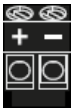
Fig. 18: Unlock the mechanical catch and remove the Touch terminal

Damage hazard!
Using excessive force to try to remove the Touch terminal from the bottom housing panel despite the catch (Lock level) could cause the mechanism to be **destroyed**.

3 Electrical connection

Connecting terminals and their function

Straightforward and largely self-explanatory wiring of components and simplified commissioning are a particularly important part of the new Systevo wiring concept. Specific functions are thus allocated to the connecting terminals via colour coding. Every function, e.g. power supply or switch output, as well as every data bus has its own colour for easier differentiation.

Function	Connecting terminal	Properties	Connection board 79CM307.. Z1 +24 V / DC BUS	Z2 24 V / DC ETH	Z3 PoE ETH	Z4 Care-View
Power in 24 V / 10 A		**Power supply** from the ward, 24 V max. 10 A	✓	✓	...	✓
2 x DBUS		**DBUS:** Use a twisted wire, max. length 200 m. Up to 4 DBUS components possible per DBUS, radial wiring possible.	✓	✓	✓	✓
BBUS		**BBUS:** Use a twisted wire, max. length 200 m. Up to 10 BBUS devices are supported, e.g. 8 bed bus modules, 2-display modules, radial wiring possible.	✓	✓	...	✓
ZBUS		**ZBUS:** Use a twisted wire, max. length 2500 m, (without repeater max. 500 m). The ZBUS supports up to 127 ZBUS devices (use repeaters after 64 devices) radial wiring **not** permitted.	✓	✓	...	✓
ABUS *		**ABUS:** Use a twisted wire, max. length 2500 m, (without repeater max. 500 m). The ABUS supports up to 127 ABUS devices (use repeaters after 64 devices) radial wiring **not** permitted.	✓	✓	✓	✓
R-ABUS *		**R-ABUS: Room audio data bus** Use a twisted wire, max. length 200 m. The R-ABUS supports up to 6 devices.	✓	✓	✓	✓
4 x NC (up) / passive I/O Down		**I/O port:** Connection for passive call units exclusively in quiescent current. **I** ➔ Call circuit input 0 V~24 V +/- 10% DC. **O** ➔ Lamp output (reassurance light) max. 100 mA (open collector acc. to GND)	✓	✓	✓	✓
3 x power for bed		**Power supply** for beds, max. 3 A (all power supply terminals for BBUS + DBUS 1 + 2 together)	✓	✓	✓	✓
ETH		Network / LAN connection	...	✓	✓	✓
ETH with PoE		Network / LAN connection (PoE)	...	...	✓	✓

✓ = available ... = not available

* For the connection boards 79CM307 Z2 / Z3 / Z4, a selection is made between ABUS / R-ABUS according to the type of termination.
The connection board 79CM307 Z1 features the ABUS and the R-ABUS.

For more information, see Chapter 3.1.

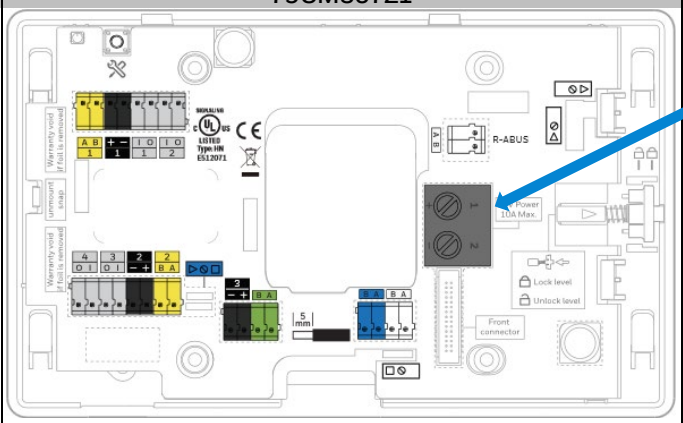
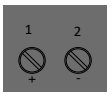
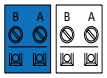
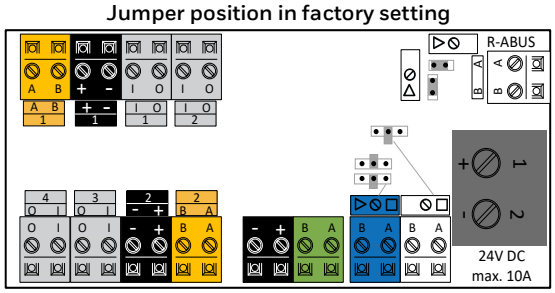
Be sure to connect with the correct polarity. Faulty or reversed polarity wiring can lead to irreversible damage.

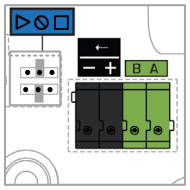
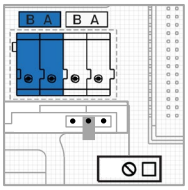
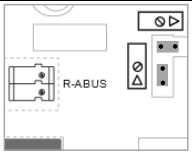
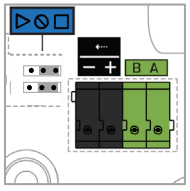
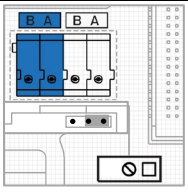
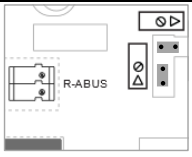
3.1 Connection board versions

Depending on the requirements or conditions of the project, different versions of the connection board can be used.
The differences lie in

- the type of power supply for the STIP
- how the data exchange between the connection boards and the call system will be established
- whether a network connection for additional services is required

3.1.1 Connection board 79CM307Z1 data bus mode

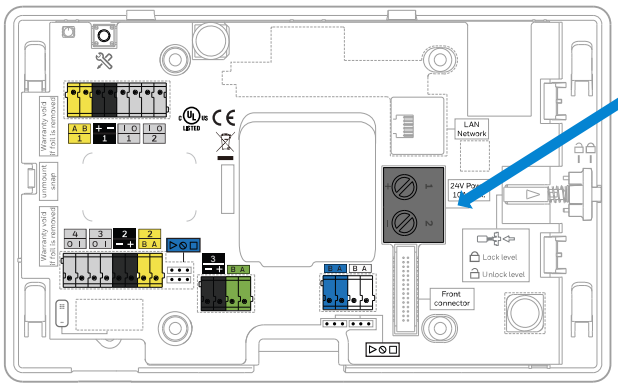
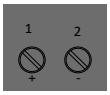
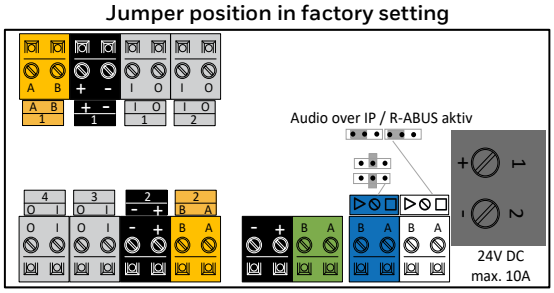
Data bus mode	Connection board 79CM307Z1	Power supply Data / audio
STIP connection board 24 V DC / BUS **Data bus mode:** In data bus mode, the power supply from the ward supply is used. The data and audio connections on the call system are established via the corridor and audio data bus. The R-ABUS enables the connection of components supported by R-ABUS in the patient room for audio transmission to the bed		 **Power supply** 24 V DC max. 10A  **ZBUS / ABUS** Connecting to the call system  **R-ABUS** Audio connection to the bed
Simplified display of the connection board STIP 24 V DC / BUS	**Jumper position in factory setting** 	

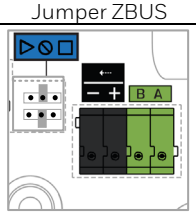
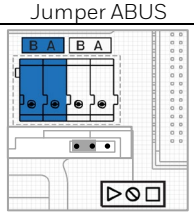
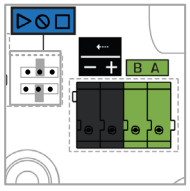
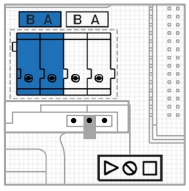
Termination of ZBUS and ABUS				
	Jumper ZBUS	Jumper ABUS	Jumper R-ABUS	
(Factory settings) Devices on the corridor and audio data bus				Device is on the corridor and audio data bus in **any** position. R-ABUS **active**.
	Jumper ZBUS	Jumper ABUS	Jumper R-ABUS	
End of corridor and audio data bus				Device terminates the **end** of corridor and audio data bus. R-ABUS **active**.

Observe the max. power for corridor components (see Chapter 10). Jumpers for R-ABUS remain plugged in at all times.

Be sure to connect with the correct polarity. Faulty or reversed polarity wiring can lead to irreversible damage.

3.1.2 Connection board 79CM307Z2 IP+ mode

IP+ mode	Connection board 79CM307Z2	Power supply Data / audio
Connection board STIP 24 V DC / ETH **IP+ mode:** In IP+ mode, the power supply from the ward supply is used. The data connection to the call system is established via the corridor data bus. The audio connection can be established via the audio bus or via the IP data connection as desired. Additional services can be made available via the network connection.		 **Power supply** 24 V DC max. 10 A  **ZBUS** Connection to the call system.  **ABUS / R-ABUS** Depending on the jumper position, either the ward audio data bus or the patient room audio data bus is applied.  **Network connection** for additional services.
Simplified display of the connection board STIP 24 V DC / ETH	Jumper position in factory setting 	

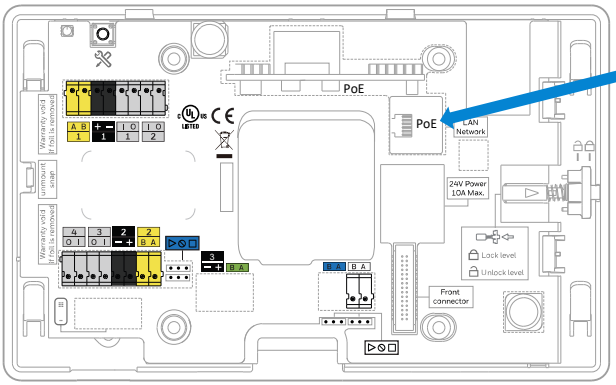
Termination of ZBUS and ABUS			
Audio connection via IP: The R-ABUS is activated when delivered. If connection is via IP, the station ABUS is **not** connected. **Data connection via ZBUS:** Devices on the ZBUS	Jumper ZBUS 	Jumper ABUS 	In this configuration, audio to the terminal **and** to the bed is possible. The ABUS terminal supplies the **R-ABUS**.
Audio connection via ABUS: For audio connection via the station ABUS, audio to the terminal is possible.	Jumper ZBUS 	Jumper ABUS 	In this configuration, audio to the terminal is possible. Device is on the corridor data bus in **any** position.

Observe the max. power for corridor components (see Chapter 10).

Be sure to connect with the correct polarity. Faulty or reversed polarity wiring can lead to irreversible damage.

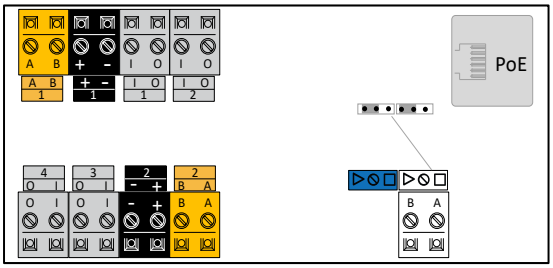
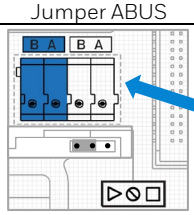
3.1.3 Connection board 79CM307Z3 IP mode

<table>
<tr><th>IP mode</th><th>Connection board 79CM307Z3</th><th>Power supply
Data / audio</th></tr>
<tr><td>Connection board STIP PoE/ETH
IP mode:
In IP mode, the power supply takes place via PoE.
The data and audio connections on the call system are established via the network connection via IP protocol. Additional services can also be made available via the network connection.

The R-ABUS enables the connection of components supported by R-ABUS in the patient room for audio transmission to the bed</td><td> 
</td><td> 

PoE voltage supply and connection to call system and additional services via network.

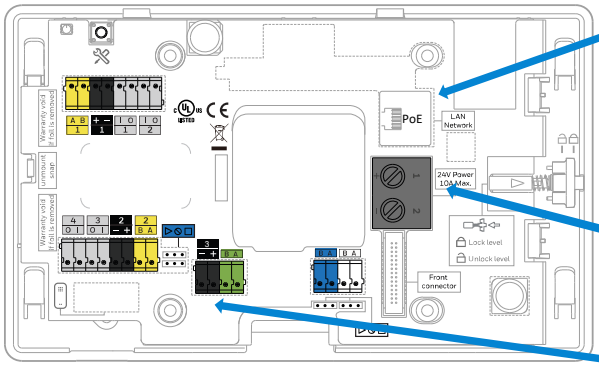
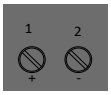
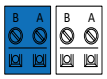
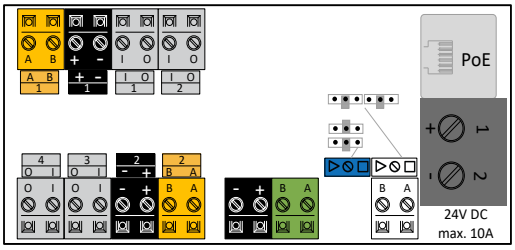
R-ABUS
To connect components supported by R-ABUS in the patient room</td></tr>
<tr><td>Simplified display of the connection board
STIP PoE / ETH</td><td>Jumper position in factory setting
 
</td><td></td></tr>
<tr><td colspan="3">Termination of ZBUS and ABUS</td></tr>
<tr><td rowspan="2">(Factory settings)
Audio connection via IP:
With this connection via IP, audio is transmitted to the terminal via IP.</td><td>Jumper ZBUS</td><td>Jumper ABUS</td></tr>
<tr><td>Not provided for this connection board.</td><td> 

The audio signal transmitted via the IP channel is made available for the corresponding components in the room via the R-ABUS.</td></tr>
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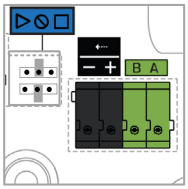
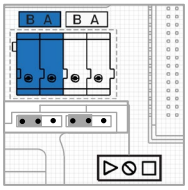
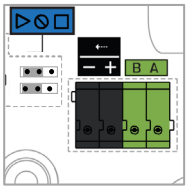
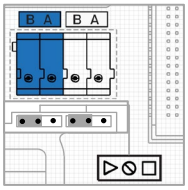
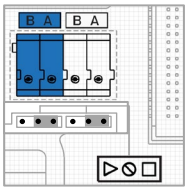
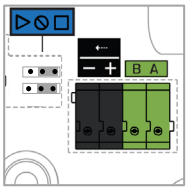
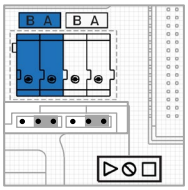
PoE as power supply is only suitable for the areas / regions in which compliance with DIN VDE 0834-1:2016-06 with regard to isolation with 2 x MOPP according to DIN EN 60601-1 is not required.
Observe max. power for corridor components if STIP is supplied via PoE (see Chapter 10).

Be sure to connect with the correct polarity. Faulty or reversed polarity wiring can lead to irreversible damage.

3.1.4 Connection board 79CM307Z4 Care View

Care View IP mode	Connection board 79CM307Z4	Power supply Data / audio
Connection board STIP / Care View IP The connection board **combines** all functions of the two connection boards **79CM307Z2 and 79CM3073** so that the power supply can take place via PoE or from the ward supply. The data and audio connections to the call system can take place via the patient room and audio data bus or via the network connection via IP. Additional services can also be made available via the network connection.		 **Network connection (with PoE** power supply) to connect to the call system and for additional services.  **Power supply** 24 V DC max. 10 A  **Power supply** 24 V DC (only for Systevo Care View IP)  **ZBUS / ABUS** Connecting to the call system
Simplified display of the connection board STIP Care View IP	**Jumper Position in Werkseinstellung** 	

Termination of ZBUS and ABUS

	Jumper ZBUS	Jumper ABUS	
Devices on the corridor data bus and audio data bus. **R-Abus active.** Audio is connected via IP. **(Factory settings)**			Device is on the corridor and audio data bus in **any** position. The R-Abus is **active**. The ABUS terminal supplies the **R-ABUS**.
Start of the corridor data bus and **R-Abus active**. Audio connected via IP.			In this configuration, audio to the terminal and to the bed is possible. The ABUS terminal supplies the **R-ABUS**.
Start of the corridor data bus and audio is connected via **ABUS**.			With this configuration, audio to the terminal is possible.
End of the corridor and audio data bus			Device terminates the **end** of corridor and audio data bus. Audio to the terminal is possible.

When using PoE as the power supply, it is **not** possible to establish safe isolation with 2 x MOPP in accordance with DIN EN 60601-1, as stipulated in DIN VDE 0834-1:2016-06. For 24 V DC power supply from the ward, a standard network connection **without** PoE must be used.

4 Interface connections

To provide a better overview and make everything easier to see and read, in the following connection diagrams the connecting wires are depicted as reaching the connecting terminals from outside.

In practice, all connecting wires will always be run from the middle out to the connecting terminals.

Cable arrangement

The depiction shows an example of a possible cable arrangement from the **middle outward** and applies to all connection board variants.

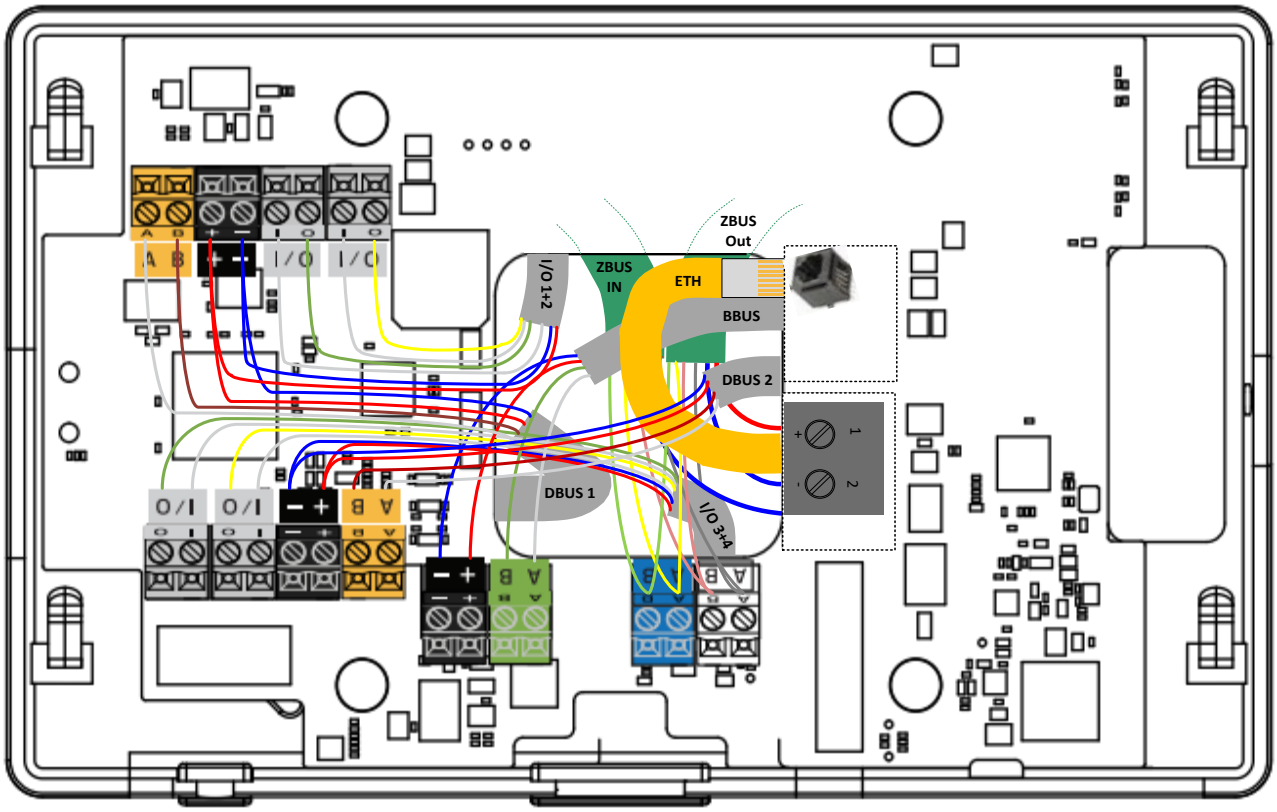


Fig. 19: Arranging connecting wires from the middle outward to the terminals

Simplified depiction

To keep the interface connections as clear as possible, a simplified depiction is used for the STIP connection boards. Only the connection options for the respective connection board are displayed. The arrangement and numbering of the connection terminals are comparable in this case.

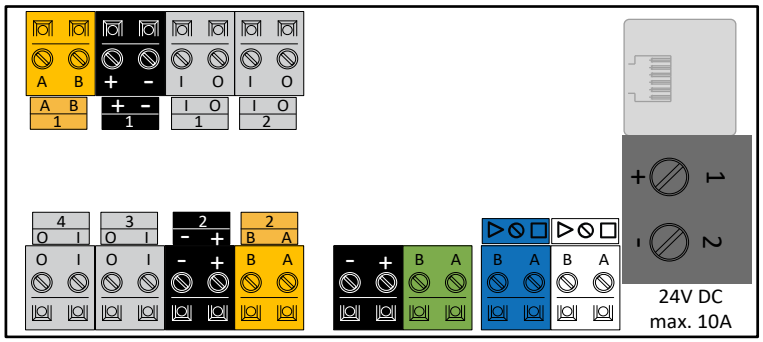
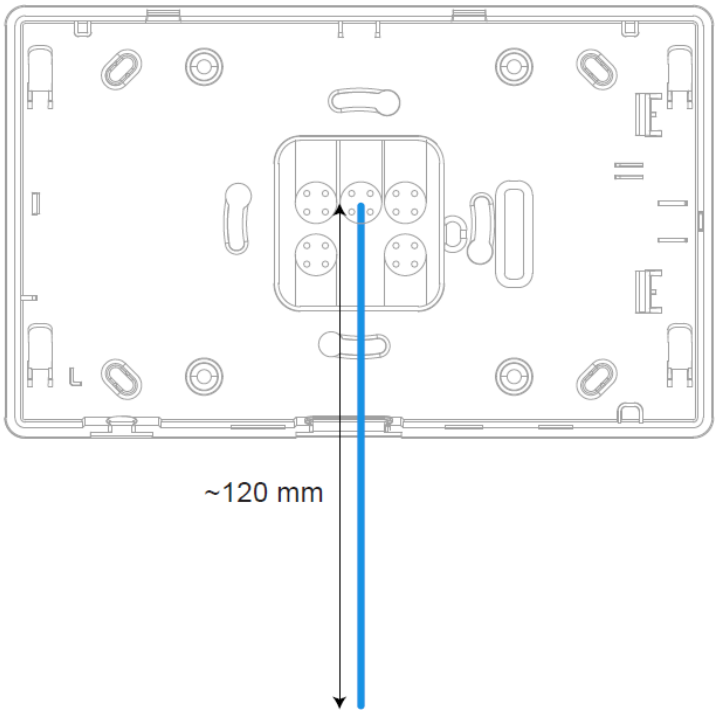


Fig. 20: Simplified depiction

4.1.1 Wiring

The space for placing the cables in the connection board is limited and thus requires a careful and space-saving cable arrangement. For this reason, cables are stripped of their outer casing so that the connecting wires – no longer than 120 mm where possible – are guided into the installation space of the connection board.



Remove the outer casing of the cable to a length of approx. 120 mm and position the cables in the rear installation socket (e.g. tunnel socket) so that only the stripped wires are placed in the connection board.

Fig. 21: Remove the outer casing of the cables

For cabling in the room, a wire diameter of 0.6 mm is recommended.

Be sure to connect with the correct polarity. Faulty or reversed polarity wiring can lead to irreversible damage.

5 Connection to the call system

Various connection boards can be used to connect the STIP to the call system. A distinction is made between the data connection via the corridor and audio data bus or the data connection via a LAN network.

5.1 Corridor and audio data bus (data bus mode / IP+ mode)

Both connection boards (STIP 24 V DC / BUS + STIP 24 V DC / ETH) are connected to the call system via the corridor/ audio data bus and supplied with power from the ward power supply.

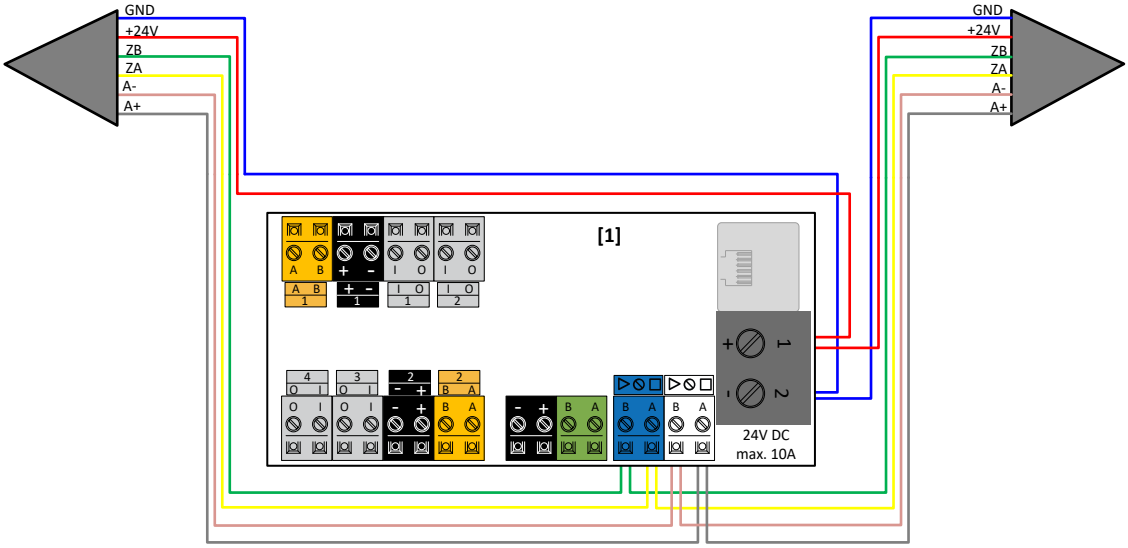


Fig. 22: Wiring ➔ Corridor and audio data bus (data bus mode / IP+ mode)

No.	Description
[1]	Connection board (79CM307Z1 / 79CM307Z2) with STIP

5.2 LAN network / PoE (IP mode)

The connection to the call system and the power supply are established via the network connection (PoE).

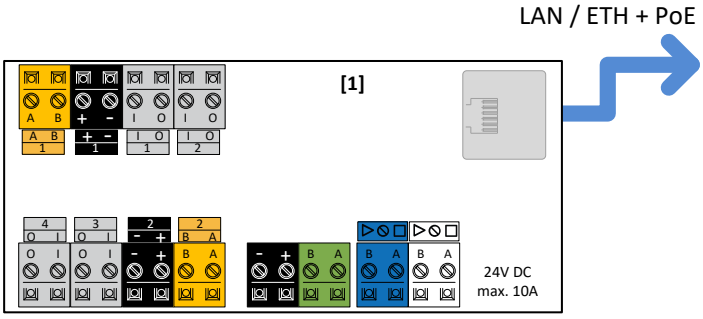


Fig. 23: Wiring ➔ LAN network / PoE (IP mode)

No.	Description
[1]	Connection board (79CM307Z3) with STIP

PoE is suitable wherever compliance with DIN VDE 0834-1:2016-06 with regard to 2 x MOPP isolation from dangerous shock currents in the patient environment is not required. This applies in particular to functional rooms and duty rooms.

6 Connection of corridor components

Use of the DBUS

The DBUS is supported from system software V12.03. The connection boards have two data bus lines. (DBUS 1 and DBUS 2) This newly developed data bus enables simple wiring in combination with simplified commissioning. Up to three DBUS components of the same type can be operated on each DBUS line, as well as one further DBUS component of a different type. Components operated on the DBUS are integrated and ready for operation in the call system without address assignment and without further configuration. If Systevo bed units of the same type are operated on one DBUS line, they are clearly identified via a slide switch on the Systevo DBUS component. Standard corridor configurations are thus easy to achieve.

Connection of DBUS components

DBUS components are interconnected with 4 wires. The flexible bus topology enables a bus and radial connection as well as a mix of the two options. The required termination is carried out by the respective component itself.

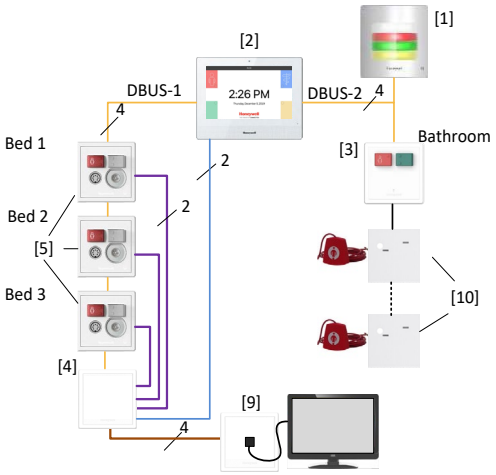
Possible DBUS components

No.	Name	Figure	Description
[1]	SCL		Systevo Corridor Lamp
[3]	SCU		Systevo Care Unit
[5]	SBU-DM2+		Systevo Bed Unit M
[6]	SBU-D2+		Systevo Bed Unit
[7]	SBU-M5		Systevo Bed Unit
[8]	SBU-M2		Systevo Bed Unit

Please note!
Depicted components may not yet be available at the time of printing.
Please check availability.

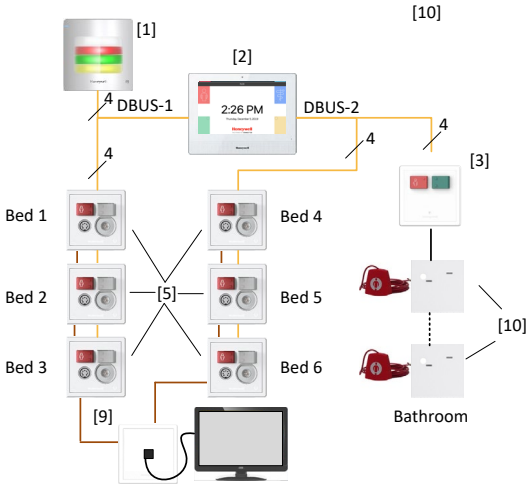
6.1 DBUS examples

3-bed room with audio & TV sound at the bed & SCU for bathroom / WC



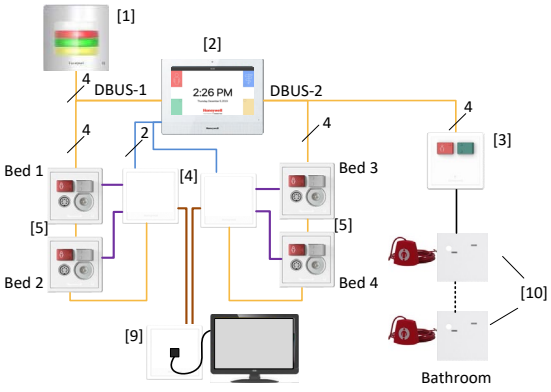
- Up to 3 beds possible
- Discreet audio to the bed
- TV sound to the bed
- Position [5] can consist of [5] to [8] as desired.
- Alternatively, the bathroom can be connected via passive I/O

6-bed room with audio at STIP & SCU for bathroom / WC



- Up to 6 beds possible
- Audio to the room via STIP
- TV sound to the bed
- Position [5] can consist of [5] to [8] as desired.
- Alternatively, the bathroom can be connected via passive I/O

4-bed room with audio & TV sound at the bed & SCU for bathroom / WC



- 4-bed solution (when using passive connections for corridor lamp and bath / WC area, up to 6 beds possible)
- Audio to the bed
- TV sound to the bed
- Position [5] can consist of [5] to [8] as desired.
- Alternatively, the bathroom can be connected via passive I/O

No.	Name	No.	Name
[1]	SCL	[6]	SBU-D2+
[2]	STIP	[7]	SBU-M5
[3]	SCU	[8]	SBU-M2
[4]	SAS	[9]	SAI
[5]	SBU-DM2+	[10]	70045A3

DBUS
R-ABUS
TV-Ton
Audio
Sonstige

Set **mode switch**. Where the **same** type of Systevo bed units are used on a DBUS line, ensure clear allocation of the DBUS ID (ID1 / ID2 / ID3).

6.2 Use of pre-assembled profiles

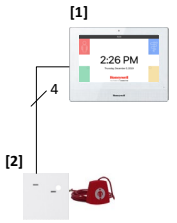
The use of pre-assembled profiles can reduce the amount of installation work and make commissioning considerably easier. For comprehensible wiring to connect passive components such as corridor lamps and passive call units, standard functions are allocated to the four I/O ports in the factory. Two pre-assembled profiles are available for these standard configurations. Both profiles can be used for 1-bed and 2-bed patient rooms with a bathroom / WC and the distinction of whether presence 2 should be displayed.
Here Profile 2 also allows the use of up to 3 DBUS components configured as passive bed at DBUS 1.
In addition, pre-assembled profile 3 enables the use of up to 3 DBUS components on the bed for each DBUS line. Up to 6 beds can be configured in this way. (Details on commissioning can be found in the Commissioning Instructions 89470J9.GB0).

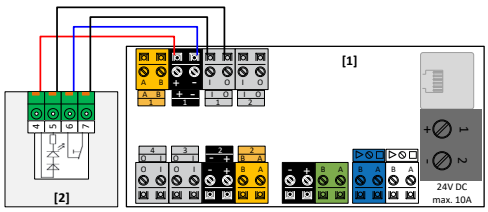
Connection of bathroom / WC components to I/O ports

The connection for BATHROOM / WC components is provided for all pre-assembled profiles. The connection of a pull cord is established when a BATHROOM / WC call is triggered. An additional cancel button resets a triggered bathroom / WC call.

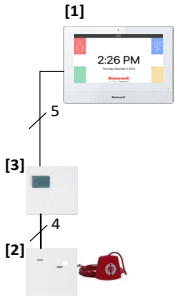
Schematic	Wiring

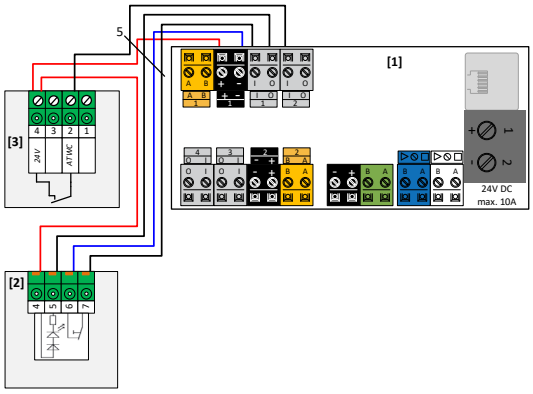
BATHROOM / WC with pull cord (70045A3) on STIP



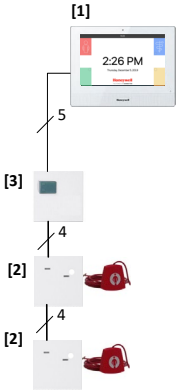


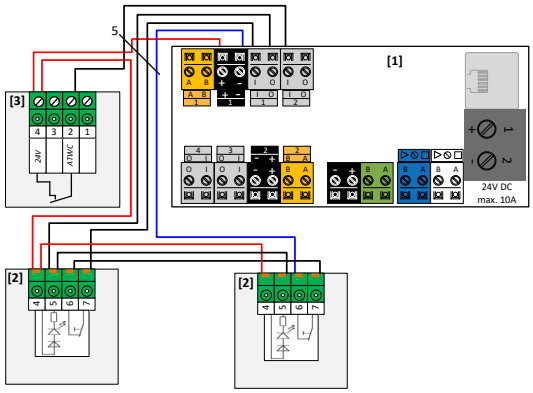
Bathroom / WC with cancel button (73642C) and pull cord (70045A3) on STIP





Bathroom / WC with cancel button (73642C) and 2x pull cord (70045A3) on STIP





No.	Description
[1]	Connection board with STIP
[2]	Pull cord 70045A3
[3]	Cancel button 73642C

6.3 Standard functions of pre-assembled profiles

Defined standard functions and the functions of dedicated connecting terminals simplify the installation and commissioning considerably, and often remove the need for costly configuration. Predefined standard functions also remain in place if data communication with the Systevo Control malfunctions or is interrupted. For this purpose, STIP is preconfigured ex factory with a defined profile (Profil2) and can therefore also be used in system environments V05.xx/V10.xx/V11.xx as a replacement for previous electronics modules. Profiles 1 + 3 can only be selected in system environments from V12.xx up. (See also chapter “Use of STIP and SCVIP in systems V5.xx / V10.xx / V11.xx” in the commissioning instructions 89470J9.GB0).

Profile-dependent standard functions:

Call trigger	**Profile 1**	**Profile 2 (Factory settings)**	**Profile 3**
	From version 12.01		**From version 12.02**
Bed 1 (with ID)	✓	✓	✓
Bed 2 (with ID)	No	✓	✓
Bed 3 (with ID)	No	No	✓
Diagnostic call to bed	No	No	✓
Diagnostic call in the corridor	No	No	✓
Call cancellation at bed	No	No	✓
Bathroom / WC with ATWC	✓	✓	✓
Plug removal call to call units 73075x and bed units SBU-Dx and SBU Mx	✓	✓ Without bed ID	✓ For SBU-Dx and SBU-Mx
Plug removal call to auto-release plug units 74199 and 74189x	✓	No, a release is signalled via hardware fault after approx. 10 seconds.	No
Lamp output			
Presence AW 1 (AL1 / green)	✓	✓	✓
Presence AW 2 (AL2 / yellow)	No	✓	✓
Call (RT / red)	✓	✓	✓
Bathroom / WC (RTWC / white)	✓	✓	✓
Reassurance light	✓	✓ Always for both beds and all corridor calls, no differentiation of the trigger location.	✓

7 Pre-defined profiles

7.1 Profile 1

This configuration type is for applications **without presence 2**, for **1 bed** and the connection of a passive corridor lamp with **3 lamp chambers** (white, red and green).

Functions:

- Bathroom / WC call and cancel button WC
- Detection of plug release with ID
- Orientation light function is controlled by the connection board. This means the orientation light function also works for plugged call units operated with passive auto-release plug units (74199 and 74189x).
- Plug release message
- No presence 2

I/O port		Description of the function
1	I	Call input bathroom / WC (RTWC)
	O	Lamp output bathroom / WC (RLWC)
2	I	Input for cancel button bathroom / WC (ATWC)
	O	Lamp output presence (AL1)
3	I	Do not use, no function (not used)
	O	Lamp output call (RL)
4	I	Call input for bed 1 (RT)
	O	Reassurance lamp output for bed call (BL)

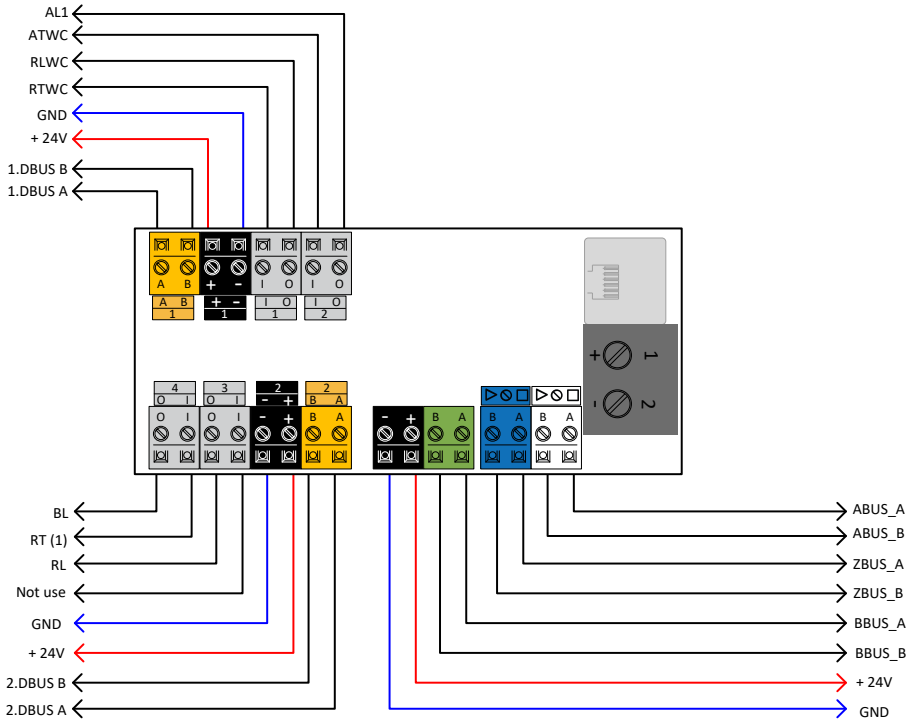
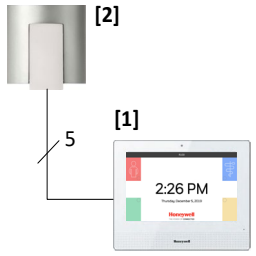
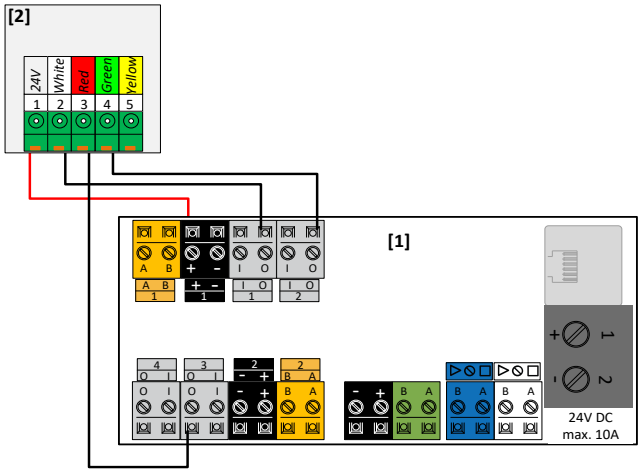
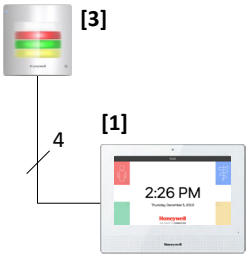
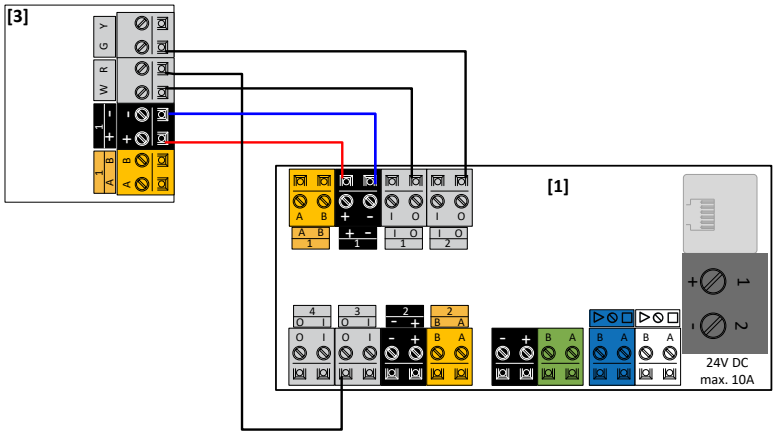


Fig. 24: Connections with their functions consistent with profile 1

7.1.1 Connection examples of passive components in profile 1

Passive corridor lamp:
By default, if **profile 1** is selected, the outputs of I/O ports 1-4 are allocated to the functions presence 1, call and BATHROOM / WC, as well as a separate output for reassurance lights and can thus activate three chambers (red, green and white) of corridor lamps.

Passive corridor lamp on STIP

Schematic	Wiring when profile 1 is selected (without presence 2)
	
	

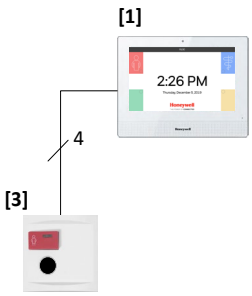
No.	Description
[1]	Connection board with STIP
[2]	Corridor lamp CL340 (passive)
[3]	Corridor lamp SCL (passive)

7.1.2 Passive call unit (73075A)

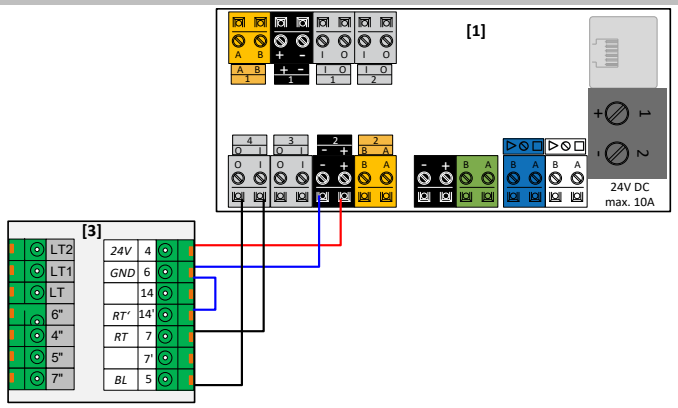
The passive I/O ports of the connection board in combination with the specified standard functions of the STIP enable the connection of existing call units.

Passive call unit (73075A) at call input for bed 1

Schematic



Wiring when profile 1 is selected



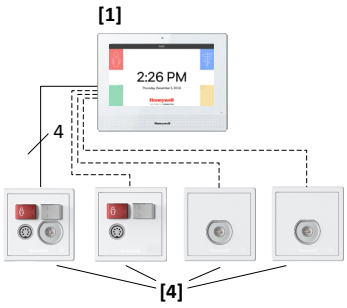
If a call is triggered at the bed, the reassurance lamp is activated in the call unit.

7.1.3 Systevo bed unit (SBU-DM2+, SBU-D2+, SBU-M2, SBU-M5) passive connection via I/O port

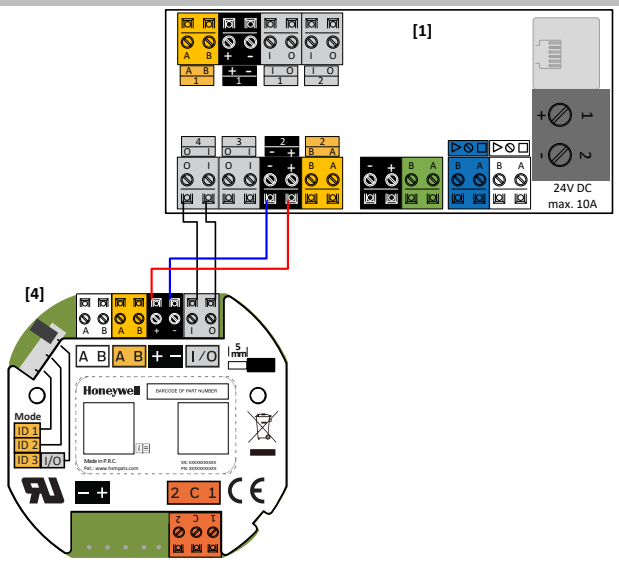
In addition to the connection via the DBUS, the Systevo bed units can also be connected via the **I/O ports** as passive units. The following connections depict example connections of a bed unit SBU-DM2+ and apply to all variants of the Systevo bed units.

Systevo bed unit (SBU-xx) at the call input for bed 1

Schematic



Wiring when profile 1 is selected



If a call is triggered at the bed, the reassurance lamp is activated in the Systevo bed unit.

Move **mode switch** on the Systevo bed unit to the ID3 / IO position.

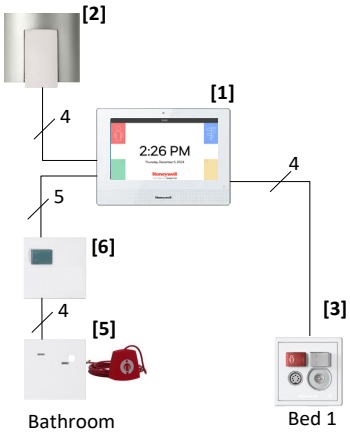
No.	Description
[1]	Connection board with STIP
[3]	Call unit 73075A
[4]	**Systevo bed unit** • SBU-D2+ • SBU-DM2+ • SBU-M2 • SBU-M5

7.1.4 Connection of passive components for a 1-bed room with bathroom / WC

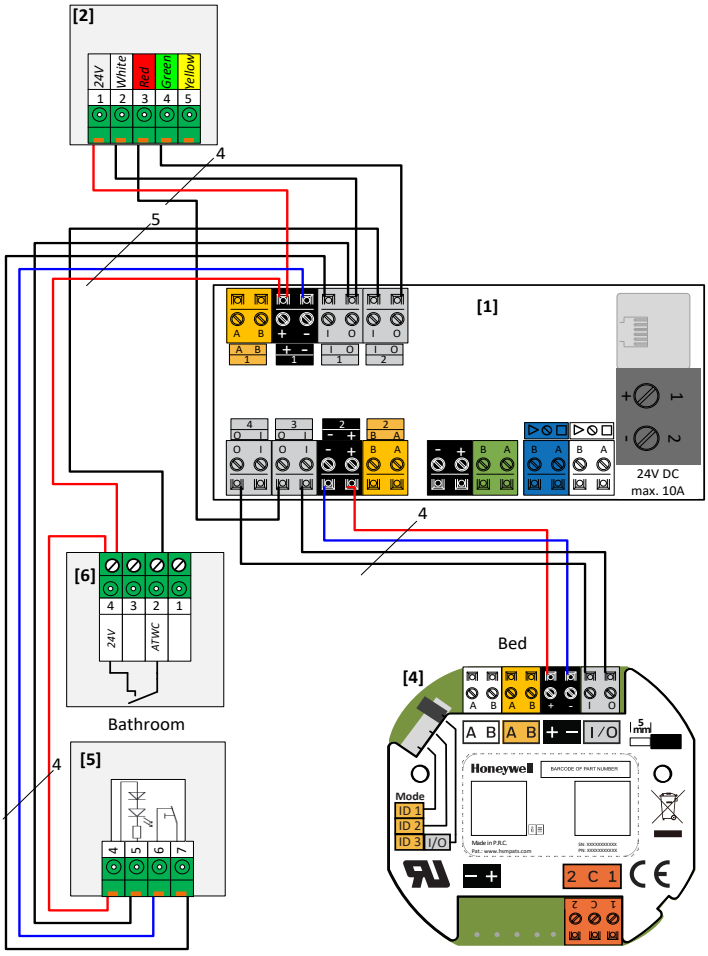
With the option to connect passive components, existing components from the BATHROOM / WC area can be adopted without changes.

A 1-bed room with bathroom / WC

Schematic



Wiring when profile 1 is selected



No.	Description
[1]	Connection board with STIP
[2]	Corridor lamp CLxx (passive)
[4]	**Systevo bed unit** • SBU-D2+ • SBU-DM2+ • SBU-M2 • SBU-M5
[5]	Pull cord 70045A3
[6]	Cancel button 73642C

7.2 Profile 2

This configuration type is for applications **with presence 2**, with up to **two beds** and the connection of a passive corridor lamp with **4 lamp chambers** (white, red, green and yellow).

Functions:

- Bed call with ID
- Bathroom / WC call and cancel button
- Detection of plug release (without ID and not for auto-release plug units 74199 and 74189x)
- Local orientation light (not for auto-release plug units 74199 and 74189x)
- Combined lamp output for all bed and corridor calls (no differentiation)
- Plug release message and orientation light function only with call units (73075xx) and bed units (SBU-Dxx and SBU-Mx).

I/O port		Description of the function
1	I	Call input bathroom / WC (RTWC)
	O	Lamp output bathroom / WC (RLWC)
2	I	Input for cancel button bathroom / WC (ATWC)
	O	Lamp output presence (AL1)
3	I	Call input for bed 2 (RT bed 2)
	O	Lamp output for all bed and corridor calls (RL+BL) (no bed ID)
4	I	Call input for bed 1 (RT)
	O	Lamp output presence 2 (AL2)

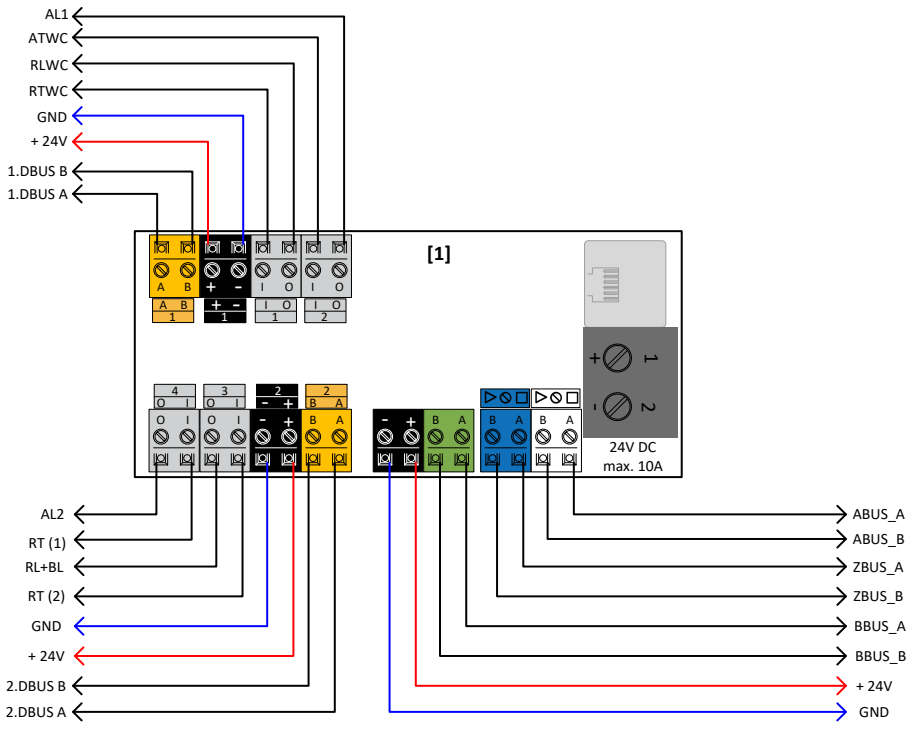


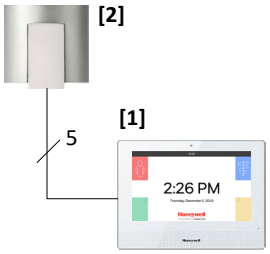
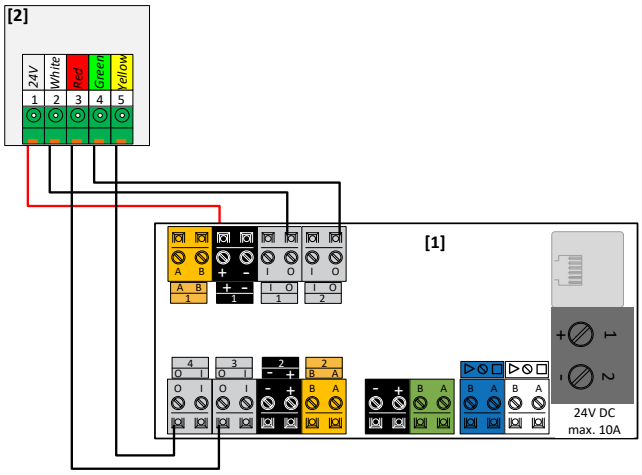
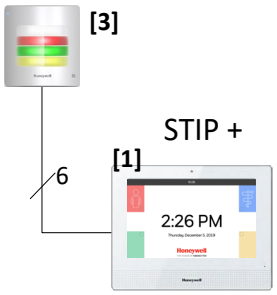
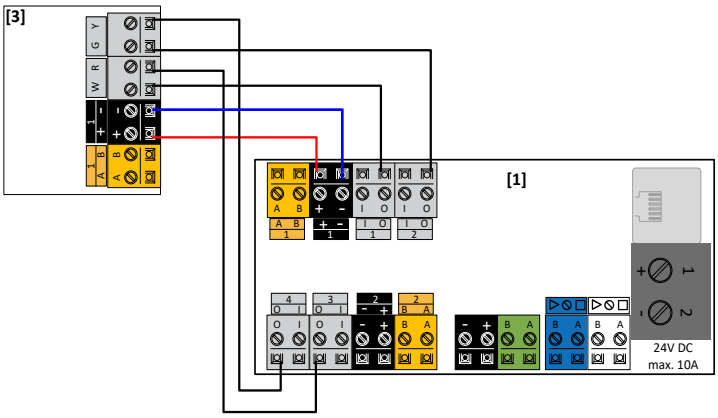
Fig. 25: Connections with their functions consistent with profile 2

7.2.1 Connection examples of passive components in profile 2

7.2.2 Passive corridor lamp

By default, if **profile 2** is selected, the outputs of I/O ports 1-4 are allocated to the functions presence 1, presence 2, call and BATHROOM / WC and can thus activate four chambers of corridor lamps.

Passive corridor lamp on STIP

Schematic	Wiring when profile 2 is selected (with presence 2)
	
	

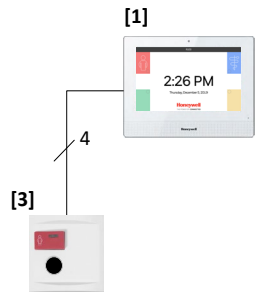
No.	Description
[1]	Connection board with STIP
[2]	Corridor lamp CL340 (passive)
[3]	Corridor lamp SCL (passive)

7.2.3 Passive call unit (73075A)

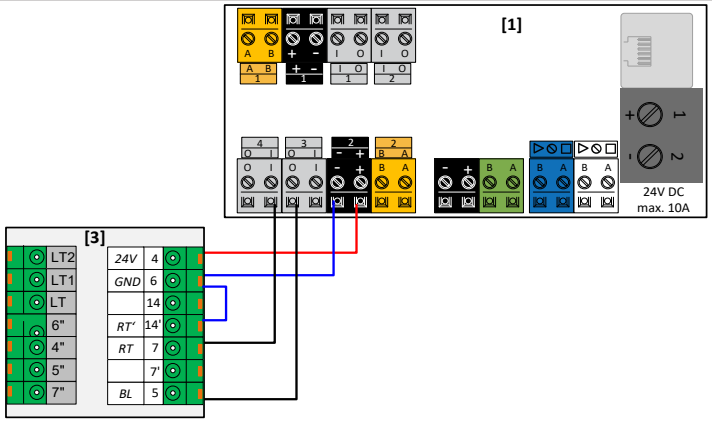
The passive I/O ports of the connection board in combination with the specified standard functions of the STIP enable the connection of existing call units.

Passive call unit (73075A) at call input for bed 1

Schematic



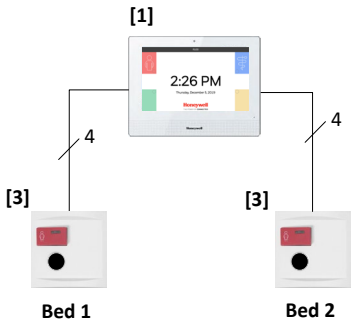
Wiring when profile 2 is selected



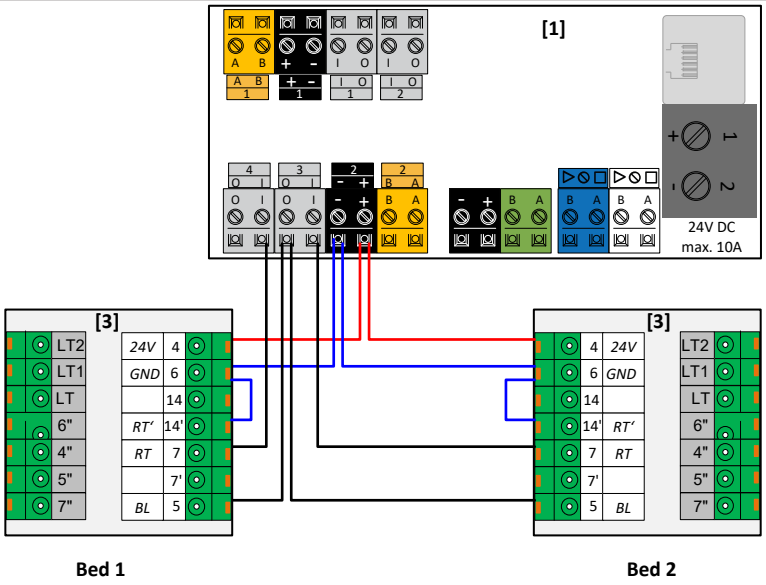
If a call is triggered in the corridor (bathroom WC / STIP) or at the bed, the reassurance lamp is activated in the call unit.

Passive call units (73075A) at call input for bed 1 and bed 2

Schematic



Wiring when profile 2 is selected



If a call is triggered in the corridor (bathroom WC / STIP) or at beds 1+ 2, the reassurance lamps are activated in the call units.

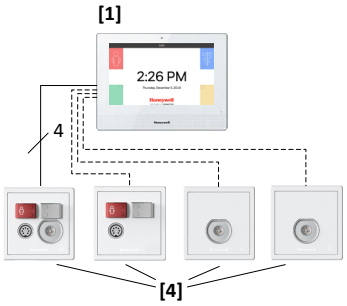
No.	Description
[1]	Connection board with STIP
[3]	Call unit 73075A

7.2.4 Systevo bed unit (SBU-DM2+, SBU-D2+, SBU-M2, SBU-M5) passive connection via I/O port

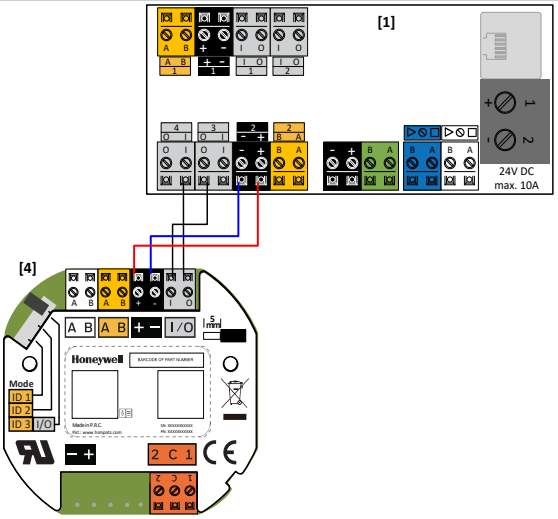
In addition to the connection via the DBUS, the Systevo bed units can also be connected via the **I/O ports** as passive units. The following connections depict example connections of a bed unit SBU-DM2+ and apply to all variants of the Systevo bed units.

Systevo bed unit (SBU-xx) at the call input for bed 1

Schematic



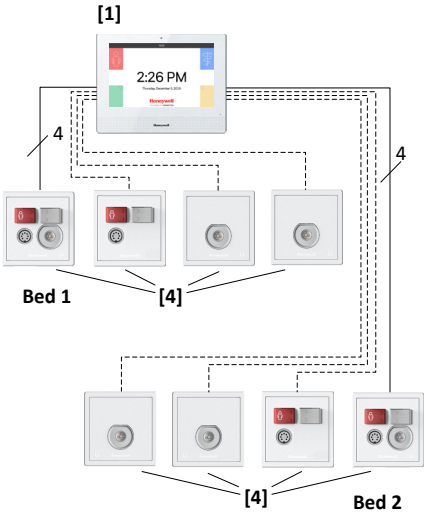
Wiring when profile 2 is selected



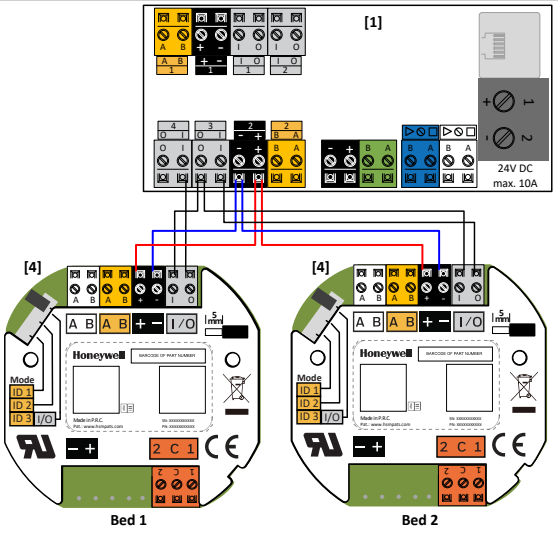
If a call is triggered in the corridor (bathroom WC / STIP) or at the bed, the reassurance lamp is activated in the Systevo bed unit.

Systevo bed units (SBU-xx) at the call input for bed 1 and bed 2

Schematic



Wiring when profile 2 is selected



If a call is triggered in the corridor (bathroom WC / STIP) or at the beds 1+ 2, the reassurance lamps are activated in the Systevo bed units.

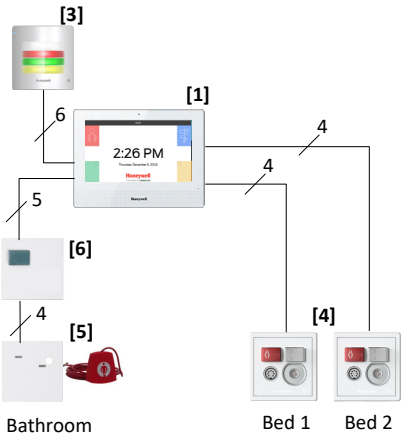
Move **mode switch** on the Systevo bed unit to the ID3 / IO position.

No.	Description
[1]	Connection board with STIP
[4]	**Systevo bed unit** • SBU-D2+ • SBU-DM2+ • SBU-M2 • SBU-M5

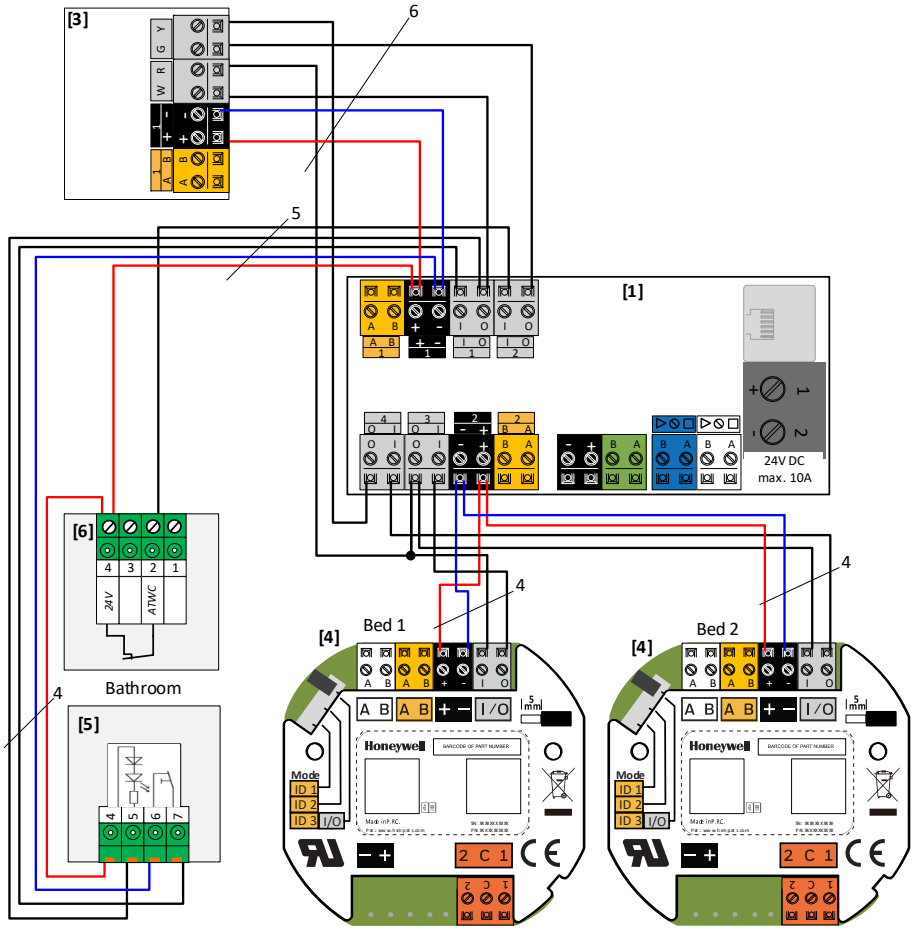
7.2.5 Connection of passive components for a 2-bed room with bathroom / WC

With the option to connect passive components, existing components from the BATHROOM / WC area can be adopted without changes.

Schematic



Wiring when profile 2 is selected



No.	Description
[1]	Connection board with STIP
[3]	Corridor lamp SCL (passive)
[4]	**Systevo bed unit** • SBU-D2+ • SBU-DM2+ • SBU-M2 • SBU-M5
[5]	Pull cord 70045A3
[6]	Cancel button 73642C

7.3 Profile 3

This configuration type is for applications **with presence 2**, with up to **three beds** at the DBUS and the connection of a passive corridor lamp with **4 lamp chambers** (white, red, green and yellow).

Functions:

- 3 components of type SBU-xx at the DBUS
- Bed call with ID
- Call cancellation at bed
- Diagnostic call to bed
- Diagnostic call in the corridor
- Plug-release detection
- Local orientation light
- Bathroom / WC call and cancel button

I/O port		Description of the function
1	I	Call input bathroom / WC (RTWC)
	O	Lamp output bathroom / WC (RLWC)
2	I	Input for cancel button bathroom / WC (ATWC)
	O	Lamp output presence (AL1)
3	I	Call input for diagnostic call in the corridor (DIA)
	O	Lamp output for all corridor calls (RL)
4	I	Input for presence 2 (AW2)
	O	Lamp output presence 2 (AL2)

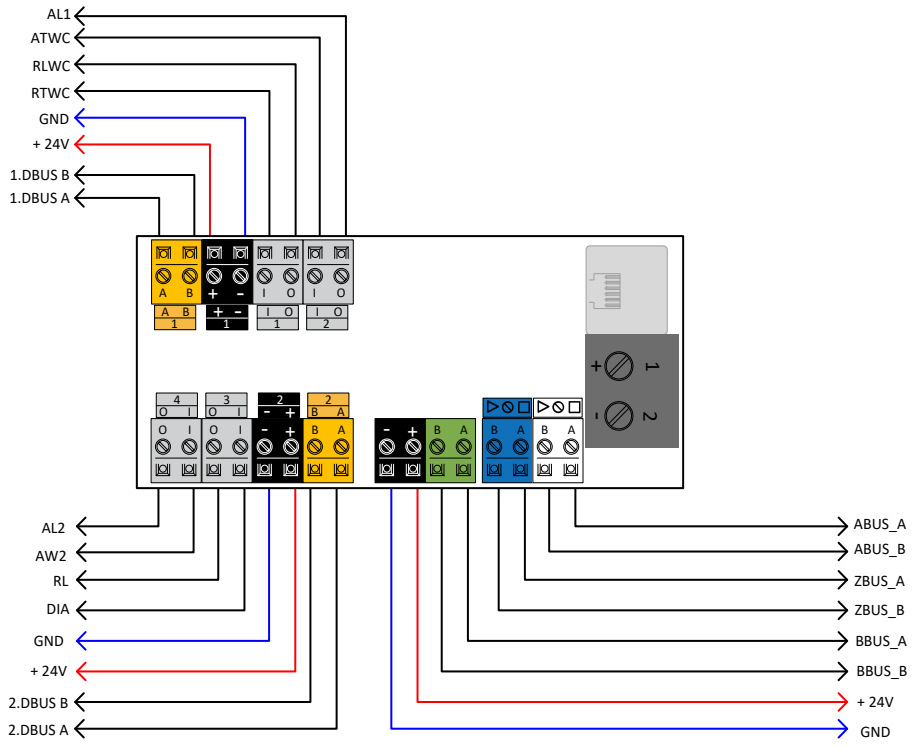


Fig. 26: Connections with their functions consistent with profile 3

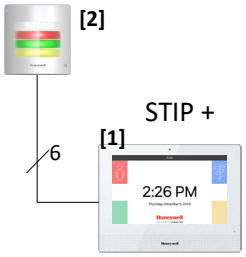
7.4 Connection examples of passive components in profile 3:

7.4.1 Passive corridor lamp:

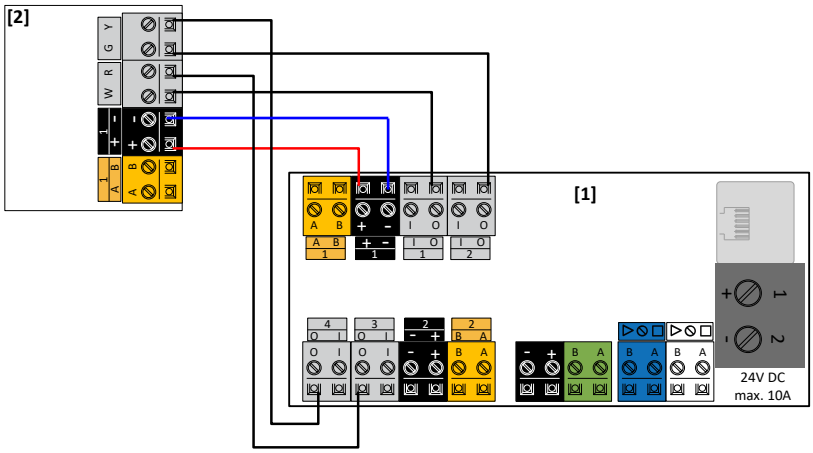
By default, if **profile 3 (as with profile 2)** is selected, the outputs of I/O ports 1-4 are allocated to the functions presence 1, presence 2, call and BATHROOM / WC and can activate four chambers of corridor lamps.

Passive corridor lamp on STIP

Schematic

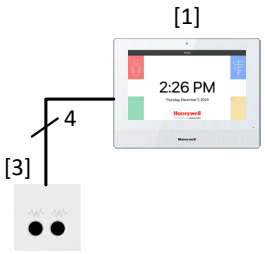


Wiring when profile 3 is selected (with presence 2)

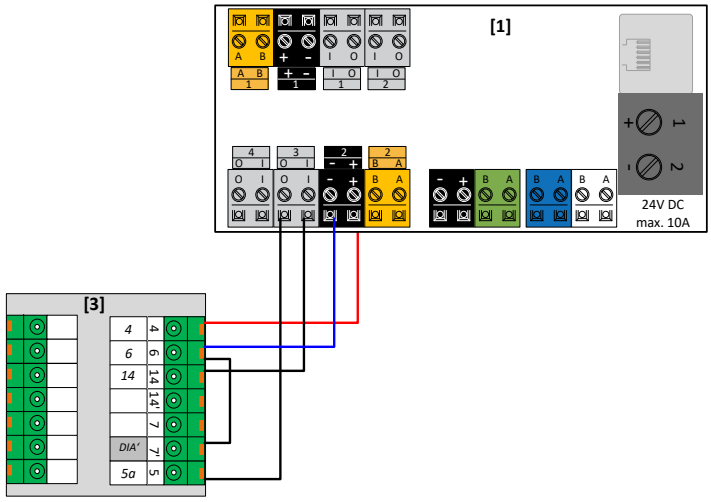


Passive call unit with 2 x 6 pole DIA socket on STIP

Schematic



Wiring when profile 3 is selected

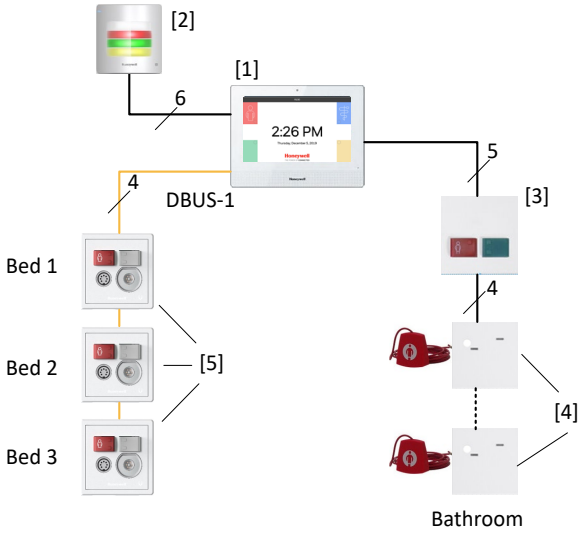


No.	Description
[1]	Connection board with STIP
[2]	Corridor lamp SCL (passive)
[3]	Passive call unit 73075D (2 x 6 pole DIA socket)

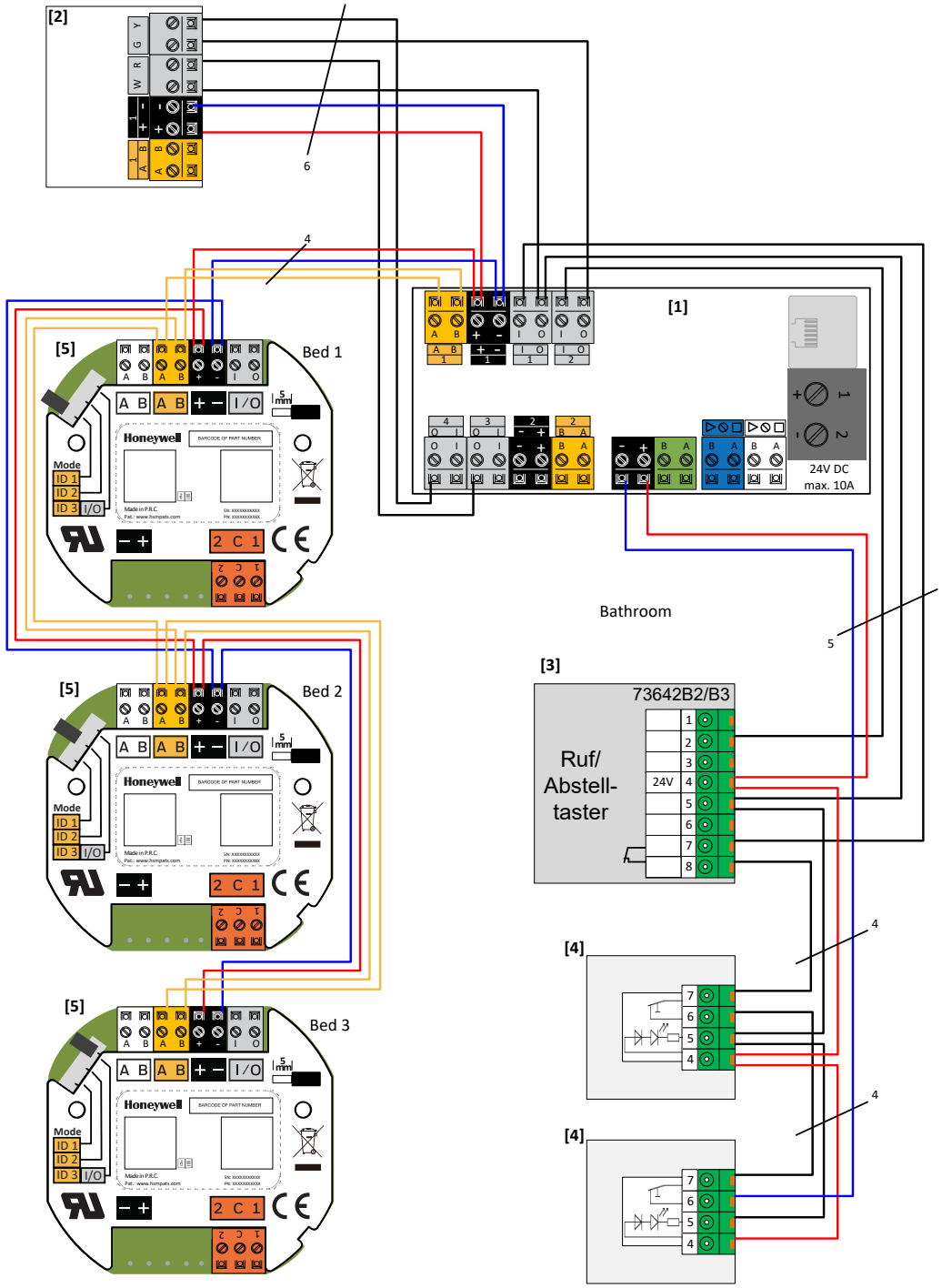
7.4.2 Connection of DBUS components for a 3-bed room with bathroom / WC

With the option to connect DBUS components, a 3-bed room with BATHROOM / WC is set up with little installation work.

Schematic



Wiring when profile 3 is selected



Set **mode switch**. Where the **same** type of Systevo bed units are used on a DBUS line, ensure clear allocation of the DBUS ID (ID1 / ID2 / ID3).

No.	Description
[1]	Connection board with STIP
[2]	Corridor lamp SCL (passive)
[3]	Call / cancel button 73642B2 / B3
[4]	Pull cord 70045A3
[5]	**Systevo bed unit** • SBU-D2+ • SBU-DM2+ • SBU-M2 • SBU-M5

When profile 3 is selected, Systevo bed units connected to the DBUS enable the connection of passive call units to trigger a DIA call to the IO port of the Systevo bed unit.

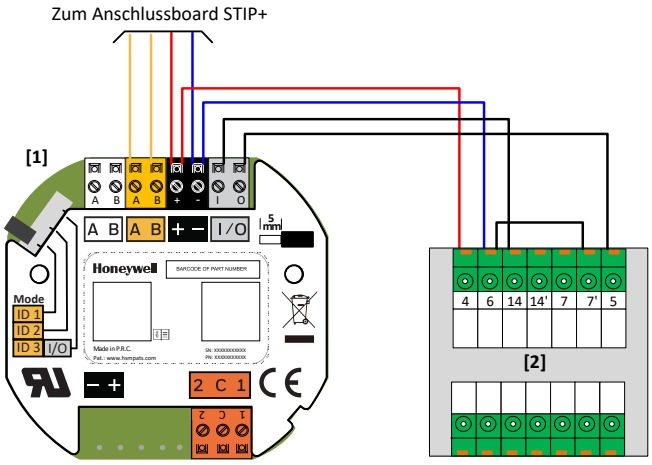
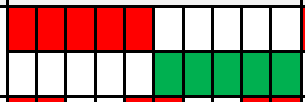
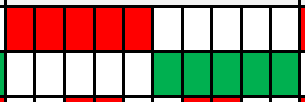
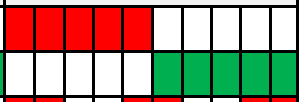
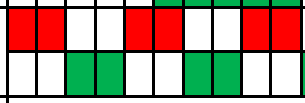
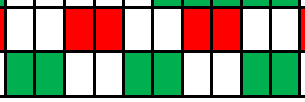
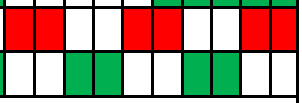


Fig. 27: Connection of a passive call unit (73075D) to the IO port of a Systevo bed unit

No.	Description
[1]	**Systevo bed unit** • SBU-D2+ • SBU-DM2+ • SBU-M2 • SBU-M5
[2]	**Passive call unit** • 73075D

8 Operating and status messages

The multi-coloured status LED on the connection board is used to display operating statuses as well as any malfunctions. Only one status can be shown at a time. In the event of several events, only the status with the highest priority will be shown (1 is the highest and 12 is the lowest priority).

Priority	Status	1 sec.	2-5 sec.	6 sec.
1	Bootloader active			
2	Firmware update active			
4	The start process and the software version are shown on the display.			
8	Fault			
9	System is operating in emergency mode			
12	Device in operation			

9 Commissioning

For further commissioning, please see the commissioning documentation 89470J9.GB0.

10 Technical specifications

General data		
Approvals/certifications	:	CE, VDE0834, UL 1069
Electrical data		
Power supply	:	24 V DC +/- 10%, PoE
Max. power for corridor components if STIP is supplied via PoE	:	285 mA (**245 mA**, if audio and RFID active)
Max. power for corridor components if STIP is supplied via 24 V DC	:	max. 2 A (all power supply terminals for BBUS + DBUS 1 + 2 together)
Average power consumption in ongoing operation @ 24 V DC	:	180 mA
Average power consumption in ongoing operation + Ethernet @ 24 V DC	:	215 mA
Maximum power consumption in ongoing operation @ 24 V DC	:	300 mA
ZBUS topology / termination / cable length	:	Bus / on board / 500 m
ABUS topology / termination / cable length	:	Bus / on board / 500 m
R-ABUS topology / termination / cable length	:	Bus + radial / on board / 200 m
BBUS topology / termination / cable length	:	Bus + radial / on board / 200 m
DBUS topology / termination / cable length	:	Bus + radial / on board / 200 m
Number of DBUS lines	:	2
Number of DBUS devices per DBUS line	:	4
I/O port: Passive input	:	0 V ~ 24 V +/- 10%
I/O port: Passive output	:	max. 100 mA open collector acc. to GND
Audio	:	Loudspeaker, 600 – 4000 Hz, 10 cm, >85 dB SPL
Interfaces	:	I/O ports: ABUS, BBUS, ZBUS, DBUS, passive IO, Ethernet port
Connection terminals (number of wires / diameter)	:	Max. three 0.6-mm wires/two 0.8-mm wires per screw terminal
Ambient conditions		
Operating temperature	:	5°C ... +40°C
Storage conditions	:	-20°C ... +70°C
Humidity	:	10% – 90% (non-condensing)
Protection class	:	IP20
Mechanical specifications		
Dimensions (H x W x D)	:	33 x 143 x 216 mm
Net weight	:	610 g
Antimicrobial / UV resistant	:	Yes
Display / resolution	:	7" TFT / (1024 x 600) 16.7 million colours
Background lighting	:	LED
Touchscreen (hardware)	:	5-point multi-touch capacitive, hardened mineral glass, hardness grade 7H
Operating button	:	Theft protection
Fire protection class	:	V-0
Housing material	:	PC + ABS-FR3010 (8% antimicrobial)
Housing colour	:	Lyric white 500N
System architecture		
Processor	:	Cortex-A53 64 Bit quad-core CPU up to 1.2 GHz
Memory	:	1 GB DDR3, 8 GB EMMC FLASH
Operating system	:	Linux long term Kernel 5.x
Wireless connections		
WIFI	:	IEEE 802.11 b/g/n radio
WIFI security	:	WEP, 802.1x, TKIP, AES, LEAP, PEAPv0, PEAPv1, EAP-M, SCHAPv2 EAP-GTC, EAP-TLS, EAP-TTLS, WPA-PSK, WPA2
Bluetooth / profile / max. range	:	Bluetooth class 4.2 / GAP, SPP, HDP, A2DP / 10 m
RFID reader	:	4 cm
Hygiene		
Disinfectants	:	70 % Isopropyl Alkohol / Sani-Cloth HB / Sani-Cloth Plus / Super Sani-Cloth / 70 % Isopropyl Alkohol Wipes / Virex 256 / CaviWipes / 409 Glass and Surface Cleaner / Windex Blue / Gentle Dish Soap and Water (Dawn) / Bacillol AF Surface disinfectant / Bacillocid rapid 0.25% Surface disinfectant / Clorox bleach 10% / DESCOSEPT AF WIPES / Sterillium (floor)

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